



Published in final edited form as:

Eur Heart J. 2026 June 02; 47(21): 2541–2560. doi:10.1093/eurheartj/ehag040.

Masters athletes with abnormal cardiovascular findings: a clinical consensus statement of the European Association of Preventive Cardiology of the ESC and the American College of Cardiology

Thijs M.H. Eijssvogels^{1,*†}, Jonathan H. Kim^{2,*†}, Vincent L. Aengevaeren³, Flavio D'Ascenzi⁴, Tim W. Churchill⁵, Elizabeth H. Dineen⁶, María Sanz-de la Garza⁷, Sabiha Gati^{8,9}, Geza Halasz¹⁰, Kristina Haugaa¹¹, Hein Heidbuchel^{12,13,14}, Andre La Gerche^{15,16,17}, Rachel Lampert¹⁸, Sheela Krishnan¹⁹, Viviana Maestrini^{20,21}, Matthew W. Martinez²², Michael Papadakis^{23,24,25}, Francesco Perone²⁶, Dermot Phelan²⁷, Sanjay Sharma^{24,28}, Paul D. Thompson²⁹, Aaron L. Baggish^{5,30,†}, Guido Claessen^{14,31,32,†}

¹Department of Medical BioSciences (route 928), Exercise Physiology Research Group, Radboud University Medical Center, P.O. Box 9101, Nijmegen, HB 6500, the Netherlands

²Emory Clinical Cardiovascular Research Institute, Department of Medicine, Emory University School of Medicine, 1750 Haygood Drive NE, HSRB II, N251, Atlanta, GA 30322, USA

³Department of Cardiology, Radboud University Medical Center, Nijmegen, the Netherlands

⁴Department of Medical Biotechnologies, Sports Cardiology and Rehab Unit, University of Siena, Siena, Italy

⁵Cardiovascular Performance Program, Division of Cardiology, Massachusetts General Hospital, Boston, MA, USA

⁶Department of Cardiovascular Diseases, Mayo Clinic Florida, Jacksonville, FL, USA

⁷Cardiovascular Institute, IDIBAPS, Hospital Clinic, Barcelona, Spain

⁸School of Medicine, Imperial College London, London, UK

⁹Department of Cardiology, Royal Brompton Hospital, London, UK

¹⁰Division of Cardiology, San Camillo Forlanini Hospital, Rome, Italy

¹¹ProCardio Center for Innovation, Department of Cardiology, Oslo University Hospital, Rikshospitalet, and University of Oslo, Oslo, Norway

¹²Department of Cardiology, Antwerp University Hospital, Antwerp, Belgium

¹³Research Group Cardiovascular Diseases, University of Antwerp, Antwerp, Belgium

*Corresponding author. Thijs.Eijssvogels@radboudumc.nl (T.M.H.E.), Jonathan.Kim@emory.edu (J.H.K.).

†Equal author contributions.

Disclosure of Interest
Nothing to declare.

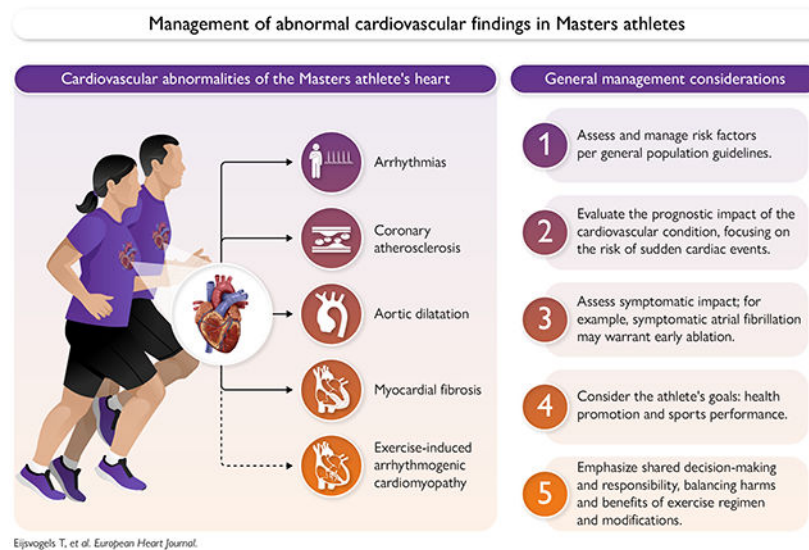
- ¹⁴Faculty of Medicine and Life Sciences, Biomedical Research Institute, LCRC, UHasselt, Hasselt, Belgium
- ¹⁵HEART Lab, St Vincent's Institute of Medical Research, Fitzroy, Australia
- ¹⁶Heart Rhythm Division, Victor Chang Cardiovascular Research Institute, Darlinghurst, Australia
- ¹⁷Department of Cardiology, St Vincent's Hospital, Fitzroy, Australia
- ¹⁸Department of Medicine, Yale School of Medicine, New Haven, CT, USA
- ¹⁹MaineHealth Cardiology, Maine Medical Center, Portland, ME, USA
- ²⁰Department of Clinical, Internal, Anesthesiology and Cardiovascular Sciences, Sapienza University of Rome, Rome, Italy
- ²¹Institute of Sports Medicine and Science, Italian National Olympic Committee, Rome, Italy
- ²²Morristown Medical Center/Atlantic Health System, Hypertrophic Cardiomyopathy Center & Sports Cardiology, Morristown, NJ, USA
- ²³City St George's, University of London, London, UK
- ²⁴St George's University Hospitals NHS Foundation Trust, London, UK
- ²⁵Cleveland Clinic, London, UK
- ²⁶Cardiac Rehabilitation Unit, Rehabilitation Clinic 'Villa delle Magnolie', Caserta, Italy
- ²⁷Gragg Center for Cardiovascular Performance, Atrium Health, Sanger Heart & Vascular Institute, Charlotte, NC, USA
- ²⁸Cardiology Clinical and Academic Group, St Georges, University of London, London, UK
- ²⁹Division of Cardiology, Hartford Hospital, Hartford, CT, USA
- ³⁰Division of Cardiology, Lausanne University Hospital, Lausanne, Vaud, Switzerland
- ³¹Hartcentrum Hasselt, Jessa Ziekenhuis, Hasselt, Belgium
- ³²Department of Cardiovascular Diseases, KU Leuven, Leuven, Belgium

Abstract

Exercise training improves cardiovascular health and reduces the risk for future cardiovascular events and mortality. However, emerging evidence suggests that Masters athletes may have a higher prevalence of cardiovascular abnormalities, such as atrial arrhythmias, coronary atherosclerosis, aortic dilatation, and myocardial fibrosis, compared to their less active peers. The clinical management of Masters athletes may be challenging as available guidelines for such conditions are generally based on data derived from symptomatic sedentary patients, limiting their applicability to highly trained individuals. Other unique challenges in the clinical assessment of Masters athletes include differences in symptomatic presentations compared to sedentary individuals, potential resistance to the initiation of pharmacologic treatment, and the increasing availability of consumer wearable health data that may provide relevant information on their cardiovascular health status. The purpose of this joint EAPC/ESC and ACC Clinical Consensus Statement is to provide an in-depth update on the current state of knowledge on

abnormal cardiovascular findings in Masters athletes. We present an expert-based approach on the diagnostic assessment, management, and prognosis of (i) atrial fibrillation, (ii) bradyarrhythmias, (iii) ventricular arrhythmias, (iv) coronary atherosclerosis, (v) aortic dilatation, (vi) myocardial fibrosis, and (vii) exercise-induced arrhythmogenic cardiomyopathy. Clinical challenges, areas of ongoing controversy, and uncertainty and the potential underlying mechanisms are discussed. We also present future perspectives and research directives to further refine current best practice strategies. This includes the need for clinical outcome studies, dedicated randomized controlled trials in athletes, and international registries with diverse populations and longitudinal follow-up to evaluate the natural history of cardiac abnormalities and facilitate development of evidence-based approaches in the clinical management of Masters athletes with cardiovascular abnormalities.

Graphical Abstract



Overview of cardiovascular abnormalities encountered in Masters athletes, including arrhythmias, coronary atherosclerosis, aortic dilatation, myocardial fibrosis, and exercise-related arrhythmogenic phenotypes, and key principles guiding their clinical management. Decision-making integrates cardiovascular risk assessment, prognostic implications, symptom burden, and the athlete's goals, emphasizing shared decision-making to balance cardiovascular safety with the health and performance benefits of lifelong exercise.

Keywords

Exercise; Cardiovascular risk; Atrial fibrillation; Myocardial fibrosis; Coronary atherosclerosis; Aortopathy; Cardiomyopathy; Athlete's heart

Introduction

Habitual physical activity and regular exercise training reduce the risk of cardiovascular disease (CVD) and all-cause mortality.^{1,2} The complex association between physical activity dose and health benefits appears to be curvilinear,³ with the largest risk reductions with

small amounts of exercise, and additional benefits with higher exercise volumes.⁴ These observations have contributed to the belief that ‘more exercise is better’. As a result, the number of people performing regular exercise training and participating in competitive sports during adulthood have markedly increased over recent decades.⁵

Masters athletes are defined as individuals aged 35 years or older that perform habitual exercise training to improve fitness levels and regularly participate in competition (Figure 1). Masters athletes’ exercise habits consistently exceed contemporary global physical activity recommendations⁶ and are primarily driven by performance or competitive goals rather than health goals. Data have shown that those who participate in organized endurance events have a lower mortality risk compared to peers from the general population with greater risk reductions for runners and cyclists compared to long-distance walkers, suggesting an important role for exercise intensity.⁷ However, high cardiorespiratory fitness does not exclude the possibility of developing cardiovascular (CV) risk factors and CVD.⁸ Emerging evidence suggests that some CV abnormalities are more prevalent in Masters athletes compared to sedentary peers.⁹ For example, individuals performing long-term exercise training have an increased prevalence of atrial arrhythmias,^{10,11} coronary calcification,¹²⁻¹⁴ and aortic dilatation.¹⁵ Furthermore, increased myocardial late gadolinium enhancement (LGE), indicative of myocardial fibrosis (MF), has been reported in lifelong athletes.¹⁶

Current guidelines for management of common CV conditions are almost exclusively based on data derived from symptomatic sedentary patients with CVD and/or sedentary asymptomatic patients with medium to high risk for adverse CV outcomes.¹⁷ Accordingly, contemporary strategies delineating diagnostic assessment, treatment options, exercise prescription and potential exercise restriction have limited generalizability to Masters athletes with abnormal CV findings. Clinical management in this context is therefore highly variable and often based on small single-centre studies, opinions, and anecdotal experiences of the treating clinician.

This Clinical Consensus Statement provides an in-depth update on the current state of knowledge on abnormal CV findings in Masters athletes (Graphical Abstract). We present an expert consensus-based approach to the diagnostic assessment, management, and prognosis of several key cardiac abnormalities observed among Masters athletes with emphasis on clinical challenges and areas of ongoing controversy and uncertainty.

Methodological approach

The writing group was assembled in alignment with the European Society of Cardiology (ESC) and the American College of Cardiology (ACC) gender policy and with efforts to include early career researchers. For the writing process, teams of four to five international experts were assigned to each content section and online sessions were organized to discuss the content and reach consensus on the considerations presented in the tables. The leadership team (T.E., J.K., A.B., and G.C.) harmonized all sections and recommendations and drafted the manuscript. The draft manuscript was subsequently circulated to all co-authors for additional suggestions and feedback, particularly directed to the clinical considerations. This process was repeated until consensus was reached. The strength of advice for clinical management considerations, as dictated by ESC policy, is summarized in Figure 2.

Clinical considerations for the masters athlete

The clinical assessment of Masters athletes aligns with routine CV practice, but a number of challenges unique to this population are noteworthy (Figure 3). First, Masters athletes often present with a chief complaint of reduced exercise capacity in isolation or in addition to common symptoms (i.e. chest pain, palpitations, etc.). This complaint raises a specific differential diagnosis inclusive of incident CVD, non-CVD systemic pathology, ageing, and physiologic detraining. It is important to quantify reductions in exercise capacity using the individual Masters athlete as his or her own point of reference, as impaired exercise capacity among Masters athletes often substantially exceed normal reference levels of the general population. The potential impact of ageing on exercise capacity deserves attention. In general, age-related declines in exercise capacity, which are inevitable, occur in a gradual fashion over years. Sudden reductions in exercise capacity are not suggestive of an ageing effect and should raise the suspicion for underlying abnormalities. Second, Masters athletes are often resistant to the initiation of pharmacologic treatment. The misconception that routine high-level exercise obviates the need for medication coupled with concerns about the impact of drugs on exercise capacity, particularly statins and agents with negative chronotropic effects (i.e. beta-blockers), are common. Nonetheless, drug therapy coupled with lifestyle modifications are often necessary to address common CV risk factors including hypertension and dyslipidemia in Masters athletes. Third, Masters athletes are increasingly using commercially available fitness devices which provide information that may be relevant to CV health. Clinicians working with Masters athletes are encouraged to develop a basic familiarity with the common metrics provided by these devices (i.e. exercise heart data, heart rate variability scores, acute workload trajectory data, etc.) to effectively integrate this potentially useful information into diagnostic assessment.¹⁸ Fourth, there are scant primary data to guide therapeutic intervention in Masters athletes. Management decisions should account for the limitations of relying on data derived from non-athletic patients and should present what is known and what is unknown in the context of shared decision-making (SDM). SDM includes an explanation of the risks and benefits of potential clinical management options and integrates the clinician's opinion about the risk-benefit balance for each potential management strategy. Finally, the use of performance enhancing drugs should be evaluated in the assessment of the Masters athlete as they may explain presenting symptoms and exacerbate the common underlying forms of abnormal CV findings discussed in this document.

Atrial fibrillation

Atrial fibrillation (AF) is an age-related arrhythmia resulting in part from longstanding underlying atrial cardiomyopathic remodelling. Masters athletes have a lower prevalence of classical AF risk factors (e.g. hypertension, obesity, or diabetes) but a higher prevalence and incidence of AF.^{10,11} Among Masters athlete Nordic skiers, the prevalence of AF is approximately twice that among comparable members of the general population.¹⁹ This is true both in men and women, although the absolute AF rates are lower in female than in male athletes.¹⁰ Adjusted hazard ratios for AF prevalence in Masters athletes are estimated to range from 2.5 to 4,²⁰ with increasing exercise duration and intensity associated with increased risk.²¹

Multiple mechanisms have been hypothesized to explain the increased incidence of AF in Masters athletes.²⁰ These include (i) structural changes in the atrium, with dilatation caused by pressure and volume overload, (ii) autonomic changes, with heightened vagal tone at rest and higher sympathetic tone during exercise, (iii) electrical changes, with slowing of conduction with increased electrical dispersion, and (iv) development of interstitial fibrosis in the atrium due to possible longstanding low grade inflammation.²² These effects of remodelling of the athlete's heart on the propensity for AF may extend beyond the active sports career. A recent study showed a four-fold higher risk for AF among former elite athletes with a high polygenic risk score.²³ On the other hand, the polygenic contribution to AF risk was similar in athletes and non-athletes (odds ratio 3.7 vs 2.0, $P = .37$). These findings emphasize the complex interplay between genetics, exercise training and AF, which may additionally be mediated through non-inherited mechanisms.²⁴

Given the increased prevalence of AF in Masters athletes, a high clinical suspicion should lead to a low threshold for diagnostic evaluation among individuals presenting with symptoms, such as palpitations, particularly among those with elevated CHA₂DS₂-VA scores. The clinical assessment of suspected AF in athletes is not different compared to non-athletes,²⁵ but treatment decisions should consider symptoms and quality of life in the context of sports performance (Table 1). A rate control strategy for symptomatic Masters athletes is often ineffective. Beta-blockers (prohibited in certain sports), calcium antagonists, and digoxin are infrequently tolerated due to side effects both at rest and during exercise.²⁶ Thus, rhythm control is usually the preferred management option. Antiarrhythmic drug therapy remains a viable option, but catheter ablation has emerged as an additional and reasonable first-line option. Prior concerns that pulmonary vein isolation (PVI) may be less efficacious in athletes than non-athletes have been disproven in several ablation series showing similar efficacy in both groups.²⁷⁻²⁹ Therefore, PVI as a first-line treatment of AF also applies to Masters athletes.^{25,30} SDM in this context remains important as complications (which still occur in 1% to 2% of PVI cases) may have an impact on athletic performance. Current data on the impact of PVI on exercise tolerance is limited and warrants further study. One retrospective study comparing exercise treadmill test results before and after PVI found no significant difference.²⁹ However, only a minority of participants underwent testing both pre- and postprocedure. Another study of highly symptomatic athletes with AF, also limited in size, showed improved exercise capacity after radiofrequency catheter ablation.²⁷ Procedural success after ablation seems comparable to non-athletes, with slow attrition rates at longer follow-up. While high volumes of endurance exercise are a well-established risk factor for AF, it remains under investigation whether prescribed reductions in exercise (detraining) leads to more effective rhythm control.

Small studies suggest that athletes with AF may have lower stroke risk than non-athletes,^{11,19} but there are no data supporting athlete-specific anticoagulation algorithms, and stroke risk assessment and prevention in athletes should follow validated risk scores such as the CHA₂DS₂-VA.^{25,31} For athletes participating in sports with risk of trauma (i.e. collision or bodily impact), SDM around continued participation is required.³² Similar to the general population, left atrial appendage occlusion may be appropriate only for Masters athletes with definitive medical contraindications to anticoagulation. There are no data to support the use of left atrial appendage occlusion for the sole purpose of

avoiding anticoagulation for Masters athletes without medical contraindications,^{25,30} so it is recommended to follow guideline-directed management.

Bradyarrhythmias

Bradyarrhythmias and marked sinus bradycardia are frequent in endurance sport Masters athletes. Sinus bradycardia with rates <40 bpm, first-degree atrioventricular block, and second-degree atrioventricular block type I may occur among healthy asymptomatic Masters athletes under resting conditions.³³ Bradyarrhythmias occur due to a combination of exercise-induced increases in vagal tone and molecular changes of ion channel proteins that lead to an attenuated sinoatrial impulse generation and conduction.^{34,35} Importantly, most athletes with sinus bradycardia and benign forms of heart block (i.e. first degree and type-1 second degree) demonstrate appropriate heart rate augmentation during exercise.

In the evaluation and management of Masters athletes, it is essential to determine whether symptoms, such as dizziness, presyncope, isolated exercise intolerance, or fatigue, are linked to clinically significant bradyarrhythmias. The diagnostic approach should include a thorough medical history, exercise testing, and ambulatory rhythm monitoring to assess the chronotropic response to exercise. While specific thresholds are currently unknown for Masters athletes, chronotropic incompetence is defined as a significantly lower than predicted heart rate during maximal effort exercise testing. Diagnostic efforts should be made to determine the relationship between maximal achieved heart rate and exercise performance. Currently, it is unknown whether reductions in sports activity for Masters athletes with chronotropic incompetence decreases the likelihood of future pacemaker implantation or if pacemaker implantation with rate-response functionality effectively enhances exercise capacity.

While bradyarrhythmias are most often asymptomatic and reversible, emerging data indicate that long-term endurance exercise may also increase the likelihood of pathological sinus node and conduction system changes.³⁶ A dose-response relationship was observed in a study of cross-country skiers with those participating in more races having a higher risk of sinus node disease or third-degree atrioventricular block.³⁷ In a follow-up study, the long-term incidence of bradycardia and the need for pacemaker implantation were, respectively, 19% and 17% more frequent in male skiers compared to the general population and most notably among the best-performing athletes and those competing in more events.³⁸ However, there is currently no evidence to support the necessity of routine long-term follow-up in these individuals in the absence of symptoms (Table 2).

Ventricular arrhythmias

The occurrence of ventricular arrhythmias in Masters athletes is generally not a normal training-related finding and requires clinical attention as this may be the first sign of an underlying cardiac disorder. Asymptomatic premature ventricular contractions (PVCs) or other ventricular arrhythmias are commonly detected in endurance athletes. Previous studies reported a prevalence of 7% in ultra-marathon finishers³⁹ and 34% among middle-aged and older athletes undergoing preparticipation stress testing.⁴⁰ The prevalence of ventricular arrhythmias on Holter monitoring does not appear to differ between Masters athletes and

the general population.⁴¹ PVCs are more likely to have higher-risk characteristics in Masters athletes compared to younger athletes,⁴⁰ but most prior studies have predominantly focused on younger athletes.⁴² The term high-risk PVCs, used to denote a series of morphologic features that increase the pretest probability of disease, has been summarized in previous clinical consensus statements for athletes^{25,32,43,44} and includes (i) PVCs with atypical morphology (i.e. those originating from locations other than the left/right ventricular outflow tract or fascicles), (ii) PVCs that are polymorphic, repetitive, or with short coupling interval (<360 ms), (iii) PVCs in combination with other abnormal ECG features for the general population, (iv) nonsuppression, new emergence of PVCs with exercise, or increased burden of PVCs during exercise, and (v) symptoms of sudden onset of exercise intolerance associated with emergence of arrhythmias. A high PVC burden in isolation, whether observed on 12-lead ECGs, exercise testing, or 24-h Holter monitoring, has been suggested as a potential high-risk marker.⁴⁵ However, it is important to note that thresholds for PVC burden are not yet clearly defined, and clear cut-points linked to clinical outcomes are unavailable in athletic cohorts, complicating the clinical interpretation and application of this finding. Nevertheless, in athletes with a PVC burden >10% by 24-h Holter monitoring, an assessment of cardiac structure and function is reasonable to exclude PVC-induced left ventricular dysfunction.

To what extent PVCs in asymptomatic Masters athletes reflect the presence of underlying disease and/or carry an unfavourable prognosis has not been well defined. While moderate exercise is protective against arrhythmias, excessive and prolonged exercise may induce structural and electrical remodelling of the heart, increasing the risk of ventricular arrhythmias.^{46,47} As such care should be taken when evaluating Masters athletes as some of the features that would be flagged as high risk in young athletes may be more prevalent in ostensibly healthy Masters athlete population.⁴⁸ Available European⁴³ and American²⁵ guidelines only provide recommendations for the younger athlete, so high-risk features need to be defined for Masters athletes in future studies. Among the Masters athlete with PVC characteristics consistent with higher risk, we suggest a clinical work-up (Table 3) including family and personal medical history, resting ECG, echocardiography, exercise testing, and 12-lead ambulatory ECG recording. This monitoring should encompass at least one or more training sessions to accurately assess the PVC burden and behaviour during physical exertion. It is currently unknown whether the burden of atypical PVCs is associated with clinical risk. Further investigations including cardiac magnetic resonance (CMR) imaging, electrophysiology studies, and genetic testing, may be appropriate on an individualized basis depending on the results of first-line testing.^{25,43,49}

Coronary atherosclerosis

Regular exercise training improves atherosclerotic risk factors, including blood pressure, inflammatory status, lipid levels, and glucose tolerance.⁵⁰ However, there is emerging evidence that coronary calcification and atherosclerosis are more common in male,^{12-14,51,52} but not female,⁵³ Masters endurance athletes compared to less active peers. A non-linear association between lifelong exercise volumes and the prevalence of a coronary artery calcium score (CACS) > 100 Agatston units (AU) has been repeatedly observed.⁵⁴ Initial studies demonstrated atherosclerotic plaques were predominantly calcified,^{12,13} suggesting

that exercise training may promote plaque stabilization by increased calcification, similar to the effect of statins.⁵⁵ In contrast, the Master@Heart study found that the prevalence of calcified plaques was similar in lifelong athletes vs controls.¹⁴ Nonetheless, the relatively high fitness status of the control group may explain this finding.⁵⁶

Mechanisms underlying coronary atherosclerosis in male Masters athletes remain uncertain. Traditional CV risk factors, such as prior tobacco use, arterial hypertension, and a family history of premature atherosclerotic disease are highly prevalent among middle-aged and older athletes^{8,57} and should be assessed in all Masters athletes.⁵⁸ Exercise physiology factors including sustained increases in blood pressure and heart rate, mechanical coronary stress, disruption of laminar intra-coronary flow, and exercise-induced inflammation may synergize with traditional CV risk factors to the development and progression of coronary calcification and atherosclerosis.⁵⁰ The magnitude of these exercise-induced mechanisms appear to be directly related to exercise intensity,⁵⁹ as suggested by data from the MARC-2 study.⁶⁰ Additional mechanisms including exercise-induced changes in lipid metabolism and calcium homeostasis,^{61,62} the consumption of an atherogenic diet, and undefined genetic factors may also contribute to the increased prevalence of coronary calcification and atherosclerosis in Masters athletes.^{50,54} Future studies are warranted to elucidate the role of these proposed pathways and to allow targeted treatment and preventive countermeasures.

Prospective outcome data in Masters athletes with coronary calcification and atherosclerosis are currently unavailable. However, findings from the Cooper Center Longitudinal Study, a general population cohort, provide important insights. Men with CACS <100 AU who engaged in high exercise volume (>3000 MET-min/week) had significantly lower all-cause mortality risk compared to less active peers with CACS <100 AU, but no differences in event rate were observed in high- vs low-volume exercisers with CACS ≥100 AU (Figure 4).⁵² In another study, those with the highest fitness levels had the lowest CV risk for each defined CACS category.⁶³ However, even among the fittest subgroup, the CV event rate was higher in those with CACS >400 AU vs CACS = 0. In summary, higher amounts of exercise and improved cardiorespiratory fitness appear to reduce the risk of coronary artery calcification^{52,63,64} but do not negate the effect of higher CACS on CV event risk.

The clinical work-up of Masters athletes with coronary calcification and/or atherosclerosis should consist of CV risk factor management, delineation of symptoms, and risk stratification (Table 4). In the absence of outcomes data in Masters athletes, CV risk factor management should be similar to the general population.¹⁷ Statins are the primary treatment for preventing CV events in athletes with elevated CACS and universal cut-off values for CACS and low-density lipoprotein goals can be used.⁶⁵ Data from the multinational CONFIRM registry revealed that individuals with CACS >300 AU face an equivalent risk of major adverse CV events compared to those with a prior diagnosis of myocardial infarction, stroke or peripheral artery disease,⁶⁶ and such risks are even higher among those with CACS >1000 AU.⁶⁷ Hence, aggressive treatment of CV risk factors should be addressed in athletes with high CACS. Similar to the general population, low-dose aspirin therapy and lipid-lowering therapy are reasonable for Masters athletes with high-risk plaque characteristics.⁶⁵

It is reasonable to utilize risk scores that integrate CACS, such as those provided by MESA⁶⁸ or Astro-CHARM,⁶⁹ to estimate incident CV events and the potential benefit from medical therapies. However, it should be recognized that available risk scores fail to capture important factors that may contribute to overall risk in Masters athletes including habitual physical activity characteristics (i.e. total exercise dose, rest/recovery patterns and annual periodization), levels of cardiorespiratory fitness, dietary macronutrient intake, and psychosocial stress. Masters athletes with coronary calcification, but no evidence of lipid-containing atherosclerosis (i.e. mixed- or low-attenuation plaques on non-invasive imaging) and no traditional CV risk factors may be seen in clinical practice. Accordingly, routine screening of CACS among low-risk Masters athletes is currently not recommended by either the ESC Sports Cardiology Guidelines or the recent American Heart Association/American College of Cardiology (AHA/ACC) Clinical Considerations for the Care of Competitive Athletes.^{17,32} For Masters athletes deemed intermediate or high-risk by traditional CV risk estimates, further clinical risk stratification is reasonable and may include assessment of CACS and plaque phenotype using computed tomographic angiography scanning.

Lifestyle is an important risk modifier for Masters athletes with coronary atherosclerosis as previous studies have shown that higher volumes of exercise and improved cardiorespiratory fitness reduce the risk of adverse health outcomes across all CACS categories compared to less trained/fit individuals.^{52,63,64} Based on current evidence, there should be no restrictions to exercise (e.g. intensity, duration, frequency) in asymptomatic athletes with high CACS in the absence of inducible myocardial ischaemia. It is imperative for Masters athletes and their medical providers to appreciate that optimal CV risk reduction is maximized at exercise levels well below the typical training volume of Masters athletes and that no amount of exercise can be considered a substitute for guideline-driven pharmacotherapy.

In asymptomatic Masters athletes deemed to be at intermediate or high risk, including those with the presence of coronary calcification, further risk stratification is reasonable. Options include exercise stress testing to evaluate for silent ischaemia and/or coronary computed tomography angiography. Exercise stress testing defines objective exercise capacity, can provoke exercise-induced arrhythmias, and can elicit ischaemic electrocardiographic ST-changes with exercise. These advantages are at least partially offset by the suboptimal sensitivity and specificity of the exercise ECG. Computed tomography angiography enables quantification of the degree, location and morphology of stenoses, but offers no information about functional capacity. In clinical practice these tests are often performed in parallel as they provide synergistic information to inform targeted risk factor management and the role of revascularization with SDM.

Symptoms of coronary artery disease (CAD) in Masters athletes may not be confined to typical angina, but can also present as unexplained declines in exercise performance or the development of exertional dyspnoea. Symptomatic Masters athletes and those with obstructive coronary atherosclerosis should be evaluated for inducible ischaemia in accordance with current guidelines. Exercise, a physiological stimulus, is preferred over pharmacological stimuli for stress testing in athletes.¹⁷

Multiple studies have demonstrated that acute thrombosis is not solely responsible for sudden cardiac arrest among men with CAD.⁷⁰⁻⁷² This suggests that stable high-grade CAD may precipitate demand-mediated ischaemic arrhythmia. The ISCHEMIA trial demonstrated that early invasive intervention has no appreciable mortality benefits over medical therapy among individuals with stable obstructive CAD.⁷³ However, it is important to note that Masters athletes are not comparable to the ISCHEMIA study population, since they may routinely exercise to intensities exceeding ischaemic threshold and are also exposed to additional precipitants of plaque destabilization including dehydration, hypercoagulability and increased coronary flow shear forces. These factors likely increase susceptibility to adverse CV events during exercise. As such and regardless of the presence of symptoms, for Masters athletes with high-risk lesions (i.e. > 50% left main stenosis, > 50% proximal left anterior descending coronary artery stenosis, two-three vessel disease with >50% stenosis and >90% stenosis in a vessel), as per ESC⁷⁴ and AHA/ACC³² guidelines, and who intend to engage in vigorous exercise, coronary revascularization can be considered after judicious SDM.¹⁷ This remains a key area for future research.

Aortic dilatation

The impact of vigorous exercise on the aorta remains an area of ongoing research. A 2013 meta-analysis revealed 3.2 mm larger aortic diameters in athletes compared to controls, suggesting a potential effect of exercise exposure on aortic dimensions.⁷⁵ Studies in young athletes (≤ 25 years old) indicate that aortic dilatation, typically defined as ≥ 40 mm in males and ≥ 34 –38 mm in females, is observed in only ~1–2% of athletes.⁷⁶⁻⁷⁹ Data among older athletic populations remain limited and report substantial variation in the prevalence of aortic dilatation. For example, mild aortic dilatation (40 to 45 mm) was present in ~20% of Masters runners and rowers,¹⁵ whereas prevalences up to 41% have been reported in former elite strength athletes, including American football⁸⁰ and rugby players⁸¹ (Figure 5). These findings suggest that sport-specific long-term exercise training, perhaps in combination with traditional risk factors, idiopathic aortic diseases, and/or prohibited substance use may contribute to increased aortic dimensions and potentially associated risk in Masters athletes.

A bicuspid aortic valve (BAV) is found in ~1% of athletes (as in the general population),⁸³ and general population data suggest that up to 50% will develop aortic enlargement.⁸⁴ Risk factors for aortic dilatation include ageing and hypertension,^{85,86} and it has been suggested that marked exercise-induced acute and transient increases in systolic blood pressure may be a risk factor for aortic dilatation.⁸² In contrast, recent studies found no impact of lifelong exercise characteristics on aortic dimensions in patients and athletes with BAV.^{87,88}

Data on sport type and aortic dilatation are inconclusive. The marked increases in systolic blood pressure that occur during strength sports and resistance training have generated concerns about aortic complications.⁸⁹ In young athletes, however, multiple studies have shown endurance athletes tend to have larger aortic dimensions compared to those in power/strength sports.⁷⁵⁻⁷⁸ Comparative studies of this sort among Masters athletes are lacking, with the exception of one analysis that found rowing (a mixed sport with power and endurance components) was associated with a higher prevalence of aortic dilatation vs running.¹⁵ Thus, the contribution of sports overall and sport type specifically to aortic

dilatation and aortic events in Masters athletes remains unclear and represents an important area of future research.

There are no data defining rates of progression among Masters athletes with aortic dilatation. Similarly, data delineating prognosis, specifically rates of acute aortic syndromes, are extremely limited, precluding any significant conclusion about the exact magnitude of risk. Return to sport after aortic surgery is also an important area of uncertainty. Very limited data ($n = 21$) suggest safety of return to endurance sport after elective surgery for BAV-associated aortic disease.⁹⁰ Finally, the risk of competitive sport participation after aneurysm repair or aortic dissection is uncertain and likely determined by surgical outcome, the presence of residual dissection and underlying disease. Participation decisions in this context should be addressed using SDM.

Masters athletes diagnosed with aortic dilatation (i.e. ≥ 40 mm for both male and female athletes, while taking body size into account) should undergo a comprehensive clinical evaluation, including assessment for features suggestive of heritable or syndromic aetiology,⁹¹ and comprehensive family history. Genetic testing may be appropriate for those at higher risk of heritable disease (Table 5).⁹¹ Masters athletes with heritable thoracic aortic disease require careful evaluation and specialist care, which is beyond the scope of this document.^{17,91}

All Masters athletes with aortic dilatation should undergo cross-sectional imaging of the entire aorta at least once. Transthoracic echocardiography is advised to assess aortic valve morphology and function. Detailed guidance for imaging in youthful competitive athletes is available, but not specific for Masters athletes.^{92,93} A customized imaging surveillance plan based on an individual athlete's diagnosis and aortic size, including side-by-side image comparison by an expert reviewer, should be developed and initiated 6–12 months after index evaluation.

Aortic dimensions relate to body size, and there is ongoing debate about how best to normalize aortic measurements. Various approaches for indexing aortic measurements have been proposed for non-athlete populations,^{94–102} but current guidelines rely primarily on the use of unadjusted aortic sizes given their association with outcomes data.⁹¹ The practical advice provided in this document aligns with this framework and does not propose separate aortic size thresholds for male and female Masters athletes, although incorporating body size into the overall risk assessment may be appropriate.

The clinical management of Masters athletes with aortic dilatation should focus on blood pressure management, exercise prescriptions, and surveillance imaging. Resting blood pressure should be measured using proper techniques,¹⁰³ and athletes should undergo ambulatory blood pressure monitoring with intermittent home surveillance and/or 24-h ambulatory monitoring. Angiotensin receptor blockers are common first-line agents, although high-quality supportive evidence is lacking in the absence of specific genetic aortopathies. While beta blocker use is common in general population patients with aortic dilatation, they are often poorly tolerated by Masters athletes. At present, there are no data to support treatment of exercise-induced hypertension if resting blood pressure is normal.

Finally, with regard to surgical intervention of Masters athletes with aortic dilatation, standard guidelines for surgical referral should be followed.^{91,104}

Limited data are available to guide decision-making about sport participation for Masters athletes with aortic dilatation. Therefore, SDM is required with the following considerations.¹⁰⁵ Further advice regarding sport participation for athletes with tricuspid or bicuspid aortic valve and without heritable aetiology is outlined in Table 5. Across all risk categories, clinical assessment should consider age, sex, and body size. Masters athletes with maximal aortic dimensions of 40–44 mm are typically considered low-risk, while those who meet established thresholds for surgical intervention are considered high-risk.^{91,104} Athletes with aortic aneurysms of 45–50 mm are at intermediate risk and require individualized evaluation. Although outcome data for this intermediate risk population are lacking, intermediate risk Masters athletes should generally be advised to avoid high-intensity resistance activity and counselled that high-intensity endurance training/competition may confer risk of progressive dilatation and incident acute aortic syndromes compared to moderate intensity. Following initial evaluation, a comprehensive longitudinal follow-up plan including serial imaging, blood pressure monitoring, and family screening should be implemented.^{91,104}

Myocardial fibrosis

Cardiac magnetic resonance imaging is emerging as a valuable tool in evaluating athletes with suspected cardiac disorders due to advanced tissue characterization, and detection of subtle myocardial abnormalities, including MF following administration of a gadolinium-based contrast agent and post-contrast imaging. MF refers to increased deposition of collagen within the extracellular matrix of the myocardium after injury, and the assessment of LGE may clarify the aetiology (i.e. ischaemic vs non-ischaemic) and severity (% of myocardium affected) of MF. Common examples of cardiac injury that precipitate MF include myocardial ischaemia, inflammation, and haemodynamic overload.¹⁰⁶ However, the identification of isolated MF among Masters athletes presents a diagnostic conundrum, as it can be challenging to distinguish between benign adaptations to athletic training and early signs of cardiomyopathy. MF may also constitute a substrate for ventricular arrhythmias due to the interplay between increased catecholamines, haemodynamic stresses during exercise, electrolyte shifts, and heterogeneous conduction in regions of MF.¹⁰⁷

The reported prevalence of MF as measured by LGE in Masters athletes ranges from 3% to 50% (Table 6).¹⁰⁸⁻¹¹⁹ Primary reasons for this variability include differences in both study design and the athletic populations studied which vary by age, sex (females underrepresented), sport, and prior intensity and duration of training. Additional study limitations include the paucity of subjects, selection bias, unknown risk factors (possible confounders), lack of appropriate controls, self-reported training histories, and unknown effects of performance enhancing and/or illicit drugs. At present, mechanisms linking intense exercise training with acquired MF are speculative but include inflammation, subclinical hypertension, demand-mediated myocardial ischaemia, and chronic ventricular volume overload.¹⁶

When MF is identified in a Masters athlete, it is important to consider the clinical context in which the CMR study was obtained and the presence of underlying CV risk factors (Table 7). All Masters athletes with significant MF should undergo evaluation for common forms cardiomyopathy that are typically associated with MF. The clinical history, pattern and extent of LGE, and pre-test probability of underlying disease inform prognosis,¹²⁰ should guide consideration of additional tests and risk stratification. For example, isolated LGE at the right ventricular insertion/hinge point(s) among asymptomatic Masters athletes is common, not associated with adverse outcomes, and should not require additional testing.^{121,122} In contrast, more extensive LGE in association with complex ventricular arrhythmias originating from affected myocardium is highly suggestive of significant pathology and should be further evaluated. The prognosis of Masters athletes with MF is dictated by the pattern and extent of LGE and the underlying aetiology.

It is reasonable to recommend temporary sports restriction in a subset of Masters athletes with LGE based on the clinical history, underlying risk factors, data from other diagnostic tests, and the pattern and extent of LGE. Requisite risk stratification testing for Masters athletes with significant MF should include sport-specific, maximal effort exercise testing and extended duration ambulatory rhythm monitoring (including while training) to assess for the presence of clinically significant ventricular arrhythmias. Treatment for Masters athletes with MF, including pharmacotherapeutics and/or defibrillator placement, should be guided by the underlying diagnosis and the presence and severity of symptoms.

Exercise-induced arrhythmogenic cardiomyopathy

Masters athletes occasionally present with ventricular arrhythmias, often with a right ventricular predominance, which can lead to the discovery of marked asymmetric ventricular dilatation during clinical work-up without clear aetiology. This clinical scenario is most often seen in Masters athletes participating in endurance sporting disciplines that have a concomitant isometric component (i.e. cycling, triathlon and rowing).⁴⁶ This uncommon clinical phenotype, for which reliable prevalence estimates are lacking and whose existence remains debated, is reported in male and female Masters athletes,¹²³ and has been termed ‘exercise-induced arrhythmogenic cardiomyopathy’ (ExI-ACM).^{46,124-126}

Initial cases of ExI-ACM were thought to represent gene-elusive variants of arrhythmogenic right ventricular cardiomyopathy, as they exhibited a highly similar phenotype but rarely carried pathogenic gene variants.^{123,127} The repetitive haemodynamic strain of pressure and volume load during exercise, which disproportionately affects the right ventricle,^{46,128} was proposed as a contributing causal mechanism. One moderate-sized study among former elite endurance athletes did not show evidence of adverse remodelling.^{119,129} Advances in cardiac genetics have recently demonstrated a polygenic contribution to extreme cardiac remodelling.¹³⁰ However, a direct link between a polygenic profile and clinically significant maladaptive remodelling has yet to be established. At present, the relative contributions of training load, genetic predisposition, and other unmeasured confounders (i.e. performance enhancing drugs, undiagnosed hypertension, subclinical myocarditis, etc.) to the development of a pathologic myocardial substrate remain uncertain.¹³¹

Data defining the prognosis of ExI-ACM are limited. An early case series of 46 high-level endurance athletes with complex ventricular arrhythmias revealed that 18 had a major arrhythmic event, including 9 athletes who died suddenly, over a median follow-up of 2 years.⁴⁷ This concerning adverse event rate occurred despite clinician-rendered advice to stop competitive sport and to undergo medical treatment, ablation, and/or implantable cardioverter-defibrillator (ICD) implantation. In this series, outcomes could not be predicted by the presence of symptoms, morphologic imaging data at the time of diagnosis, or the results of non-invasive arrhythmia evaluation. The only predictive factor of adverse events was inducibility of sustained ventricular tachycardia or ventricular fibrillation during invasive electrophysiological testing. This initial case series, accrued from three tertiary care sports cardiology centres, likely identified the most severe phenotypes with the worse prognoses.⁴⁷

Given the ongoing uncertainties, diagnosing ExI-ACM remains challenging and is primarily a diagnosis of exclusion. Before making this determination, all other potential causes of structural cardiomyopathy and electrical disease must be first carefully considered. Key differential diagnoses to exclude are genetic arrhythmogenic cardiomyopathy,¹³² non-dilated left ventricular cardiomyopathy, dilated cardiomyopathy, and other arrhythmogenic conditions such as sarcoidosis.¹²⁰ When evaluating athletes for suspected ExI-ACM, attention should focus on high-risk structural and functional features that may impact return-to-play decisions, since continued unrestricted exercise can facilitate progression of the pathogenic substrate.^{123,133} These features include disproportionate right ventricular enlargement relative to left ventricular size, reduced right ventricular ejection fraction (typically <40% on CMR; values <45% may still be physiological in ~1 in 6 elite young athletes),¹³⁰ and impaired right ventricular free wall strain, with values less negative than -20% indicating dysfunction.¹³⁴ Right ventricular wall motion abnormalities (akinesia or dyskinesia) are never adaptive in athletes and should always be considered pathological,¹³⁵ though they are uncommon in ExI-ACM¹³⁶ and more characteristic of familial arrhythmogenic cardiomyopathy, where they represent a major diagnostic criterion.

The management of Masters athletes with suspected ExI-ACM is summarized in Table 8 in alignment with the general ESC guidelines for cardiomyopathies.¹²⁰ Several observational studies suggest that exercise detraining may contribute to the partial reversal of cardiac structural changes and a reduction of ventricular arrhythmias.²⁵ Prescribed detraining for athletes with suspected ExI-ACM is a reasonable approach for those without sustained or life-threatening arrhythmias and should be coupled with serial re-evaluation to guide risk stratification and clinical management decisions.²⁵ Management of arrhythmias, whether by medication, ablation or ICD implantation should conform to the latest evidence and recommendations on ventricular arrhythmias and the prevention of sudden cardiac death in patients with cardiomyopathies.^{25,120,137} In athletes with an initial indication for ICD implantation, reassessment after a detraining period may be appropriate. The wearable cardioverter-defibrillator could be proposed as an interim solution in those judged to be at high risk, during exercise restriction.^{137,138} Epicardial ablation may be effective in athletes with documented ventricular tachycardia and isolated subepicardial right ventricular outflow tract scar.¹²⁵ Finally, athletes with suspected ExI-ACM who develop right ventricular

failure and/or secondary tricuspid regurgitation,¹³⁹ should be managed in accordance with contemporary heart failure guidelines.¹⁴⁰

Future perspectives and research directions

Over the past two decades, evidence on CV abnormalities in Masters athletes has evolved, yet significant uncertainties persist. Best practices for managing those with CV risk or CVD lack strong evidence and rely on expert consensus and extrapolation from the general population—despite key differences. First, the ‘exercise paradox’, referring to increased acute cardiac risk during exercise even in highly fit individuals, remains imprecisely quantified in Masters athletes.²⁷ Additionally, the risk-benefit balance between CV risk factors such as coronary artery calcification and the protective effects of high cardiorespiratory fitness is unclear.^{63,141} Finally, the mechanisms underlying potentially maladaptive cardiac phenotypes in Masters athletes remain incompletely understood.^{9,50,142}

Addressing these uncertainties will require rigorously controlled, longitudinal studies. Future Masters athlete registries should prioritize diverse populations with balanced male and female representation and initiate outcomes tracking as early as possible in life. Comprehensive phenotyping, inclusive of serial clinical assessments, 12-lead ECG, advanced imaging, and laboratory assessments data, is essential. Accurate prospective and objective measurement of exercise dose (duration, intensity, and frequency) and sporting discipline should replace reliance on retrospective recall. In addition, factors unique to this population, such as dietary intake (i.e. nutritional supplements/diet during high-training volume periods and competition), and psychosocial stress should be examined alongside traditional CV risk assessments. Future studies should also take the use of performance enhancing and/or illicit drug use into account, as this is often unreported despite its known deleterious CV effects.¹⁴³ Notably, no studies have explored social determinants of health or potential racial disparities in health outcomes among Masters athletes, whereas sex-based disparities in the incidence of disease among Masters athletes remain understudied and incompletely understood, highlighting a critical gap for future research.¹⁴⁴ Finally, advancements in artificial intelligence and computational cardiology may further inform this work.

Longitudinal registry data are required to better understand the potentially maladaptive clinical findings in Masters athletes. Differentiation of stable coronary calcifications vs unstable mixed-morphology atherosclerotic plaque, including their underlying mechanisms and clinical outcomes, will improve clinical management. Similarly, controlled registry data will help clarify causal mechanisms and distinguish pathological from physiological adaptations in AF, subclinical MF, and aortic dilatation. The possibility of ExI-ACM also warrants further investigation. While credible prevalence estimates of this proposed pathology are lacking, it appears to be a rare condition affecting a small subset of Masters athletes. Identifying factors that determine host susceptibility, including but not limited to genetic influences, is needed. Emerging evidence suggests that up to ~50% of gene-negative arrhythmogenic cardiomyopathy cases may harbor previously unrecognized genetic mutations,¹⁴⁵ suggesting an important but still poorly understood genetic component in this pathologic phenotype.

To date, no clinical trials or randomized controlled trials have focused exclusively on athletes, including Masters athletes. Accordingly, expert consensus opinion, based largely on observational data and clinical experience, drive current practice patterns and may deviate from general population guidelines. For example, consideration of coronary revascularization has been recommended for Masters athletes with stable obstructive CAD in ESC Sports Cardiology guidelines and AHA/ACC scientific statements.^{17,32} In addition, sinus rhythm maintenance for Masters athletes with AF, which increasingly involves first-line PVI, remains complex and incompletely understood. Key gaps include understanding the effects of detraining on AF burden, the risk of AF recurrence with resumed high-intensity training post-PVI, the long-term impact of PVI on exercise capacity, and the efficacy of emerging ablation techniques (e.g. pulsed field ablation) in highly active individuals. While conducting randomized controlled trials in Masters athletes presents challenges, such studies are essential to establishing data-driven best practices for this unique population facing the challenges of CVD.

Conclusions

The global population of Masters athletes continues to rise. While Masters athletes benefit from high cardiorespiratory fitness and reduced cardiac mortality, they often harbour traditional CV risk factors and increased risk of common forms of CV abnormalities. Despite recent advances in our understanding of CVD among Masters athletes, uncertainties remain. Cardiac care for Masters athletes should include comprehensive traditional CV risk factor assessment, lifestyle counselling, and medical therapy when appropriate (Graphical Abstract). Key considerations that should guide therapeutic interventions include symptom burden and their impact on quality of life, the Masters athletes' preferences and goals, and disease-specific prognosis on a case-by-case basis. SDM represents an essential approach in determining treatment strategies and exercise prescription. Future research is warranted to more clearly delineate the benefits and potential risks of high-dose exercise in Masters athletes with and without established CV abnormalities.

Supplementary Material

Refer to Web version on PubMed Central for supplementary material.

Funding

T.M.H.E. was supported by an Established Investigator E-Dekker grant (#03-002-2023-0036) and FIT-HEART consortium grant (#01-001-2024-0621) of the Dutch Heart Foundation.

Data Availability

No data were generated or analysed for or in support of this paper.

References

1. Franklin BA, Eijsvogels TMH, Pandey A, Quindry J, Toth PP. Physical activity, cardiorespiratory fitness, and cardiovascular health: a clinical practice statement of the ASPC part I: bioenergetics, contemporary physical activity recommendations, benefits, risks, extreme exercise regimens,

- potential maladaptations. *Am J Prev Cardiol* 2022;12:100424. 10.1016/j.ajpc.2022.100424 [PubMed: 36281324]
2. Franklin BA, Thompson PD, Al-Zaiti SS, Albert CM, Hivert MF, Levine BD, et al. Exercise-related acute cardiovascular events and potential deleterious adaptations following long-term exercise training: placing the risks into perspective-an update: a scientific statement from the American Heart Association. *Circulation* 2020;141:e705–36. 10.1161/CIR.0000000000000749 [PubMed: 32100573]
 3. Eijsvogels TM, Molossi S, Lee DC, Emery MS, Thompson PD. Exercise at the extremes: the amount of exercise to reduce cardiovascular events. *J Am Coll Cardiol* 2016;67:316–29. 10.1016/j.jacc.2015.11.034 [PubMed: 26796398]
 4. Stens NA, Bakker EA, Mañas A, Buffart LM, Ortega FB, Lee DC, et al. Relationship of daily step counts to all-cause mortality and cardiovascular events. *J Am Coll Cardiol* 2023;82:1483–94. 10.1016/j.jacc.2023.07.029 [PubMed: 37676198]
 5. Vitti A, Nikolaidis PT, Villiger E, Onywera V, Knechtle B. The “New York City Marathon”: participation and performance trends of 1.2 M runners during half-century. *Res Sports Med* 2020;28:121–37. 10.1080/15438627.2019.1586705 [PubMed: 30889965]
 6. Bull FC, Al-Ansari SS, Biddle S, Borodulin K, Buman MP, Cardon G, et al. World Health Organization 2020 guidelines on physical activity and sedentary behaviour. *Br J Sports Med* 2020;54:1451–62. 10.1136/bjsports-2020-102955 [PubMed: 33239350]
 7. Bakker EA, Aengevaeren VL, Lee DC, Thompson PD, Eijsvogels TMH. All-cause mortality risks among participants in mass-participation sporting events. *Br J Sports Med* 2024;58:421–6. 10.1136/bjsports-2023-107190 [PubMed: 38316539]
 8. Shapero K, Deluca J, Contursi M, Wasfy M, Weiner RB, Lewis GD, et al. Cardiovascular risk and disease among masters endurance athletes: insights from the Boston MASTER (masters athletes survey to evaluate risk) initiative. *Sports Med Open* 2016;2:29. 10.1186/s40798-016-0053-0 [PubMed: 27547715]
 9. Thompson PD, Eijsvogels TMH, Kim JH. Can the heart get an overuse sports injury? *NEJM Evid* 2023;2:EVIDra2200175. 10.1056/EVIDra2200175 [PubMed: 38320102]
 10. Drca N, Larsson SC, Grannas D, Jensen-Urstad M. Elite female endurance athletes are at increased risk of atrial fibrillation compared to the general population: a matched cohort study. *Br J Sports Med* 2023;57:1175–9. 10.1136/bjsports-2022-106035 [PubMed: 37433586]
 11. Svedberg N, Sundström J, James S, Hållmarker U, Hambraeus K, Andersen K. Long-term incidence of atrial fibrillation and stroke among cross-country skiers. *Circulation* 2019;140:910–20. 10.1161/CIRCULATIONAHA.118.039461 [PubMed: 31446766]
 12. Aengevaeren VL, Mosterd A, Braber TL, Prakken NHJ, Doevendans PA, Grobbee DE, et al. Relationship between lifelong exercise volume and coronary atherosclerosis in athletes. *Circulation* 2017;136:138–48. 10.1161/CIRCULATIONAHA.117.027834 [PubMed: 28450347]
 13. Merghani A, Maestrini V, Rosmini S, Cox AT, Dhutia H, Bastiaenen R, et al. Prevalence of subclinical coronary artery disease in masters endurance athletes with a low atherosclerotic risk profile. *Circulation* 2017;136:126–37. 10.1161/CIRCULATIONAHA.116.026964 [PubMed: 28465287]
 14. De Bosscher R, Dausin C, Claus P, Bogaert J, Dymarkowski S, Goetschalckx K, et al. Lifelong endurance exercise and its relation with coronary atherosclerosis. *Eur Heart J* 2023;26:2388–99. 10.1093/eurheartj/ehad152
 15. Churchill TW, Groezinger E, Kim JH, Loomer G, Guseh JS, Wasfy MM, et al. Association of ascending aortic dilatation and long-term endurance exercise. *JAMA Cardiol* 2020;5:522. 10.1001/jamacardio.2020.0054 [PubMed: 32101252]
 16. van de Schoor FR, Aengevaeren VL, Hopman MT, Oxborough DL, George KP, Thompson PD, et al. Myocardial fibrosis in athletes. *Mayo Clin Proc* 2016;91:1617–31. 10.1016/j.jmayocp.2016.07.012 [PubMed: 27720455]
 17. Pelliccia A, Sharma S, Gati S, Bäck M, Börjesson M, Caselli S, et al. 2020 ESC guidelines on sports cardiology and exercise in patients with cardiovascular disease. *Eur Heart J* 2021;42:17–96. 10.1093/eurheartj/ehaa605 [PubMed: 32860412]

18. Petek BJ, Al-Alusi MA, Moulson N, Grant AJ, Besson C, Guseh JS, et al. Consumer wearable health and fitness technology in cardiovascular medicine: JACC state-of-the-art review. *J Am Coll Cardiol* 2023;82:245–64. 10.1016/j.jacc.2023.04.054 [PubMed: 37438010]
19. Johansen KR, Ranhoff AH, Sørensen E, Nes BM, Heitmann KA, Apelland T, et al. Risk of atrial fibrillation and stroke among older men exposed to prolonged endurance sport practice: a 10-year follow-up. The Birkebeiner ageing study and the Tromsø study. *Open Heart* 2022;9:e002154. 10.1136/openhrt-2022-002154 [PubMed: 36396296]
20. Flannery MD, Kalman JM, Sanders P, La Gerche A. State of the art review: atrial fibrillation in athletes. *Heart Lung Circ* 2017;26:983–9. 10.1016/j.hlc.2017.05.132 [PubMed: 28606607]
21. Boraita A, Santos-Lozano A, Heras ME, González-Amigo F, López-Ortiz S, Villacastín JP, et al. Incidence of atrial fibrillation in elite athletes. *JAMA Cardiol* 2018;3:1200–5. 10.1001/jamacardio.2018.3482 [PubMed: 30383155]
22. Guasch E, Benito B, Qi X, Cifelli C, Naud P, Shi Y, et al. Atrial fibrillation promotion by endurance exercise: demonstration and mechanistic exploration in an animal model. *J Am Coll Cardiol* 2013;62:68–77. 10.1016/j.jacc.2013.01.091 [PubMed: 23583240]
23. Flannery MD, Canovas R, Janssens K, Mitchell AM, D'Ambrosio P, Spencer LW, et al. Atrial fibrillation in former world-class rowers: role of environmental and genetic factors. *Eur Heart J* 2025;46:5114–25. 10.1093/eurheartj/ehaf369 [PubMed: 40561495]
24. Fatkin D, Cox CD, Huttner IG, Martinac B. Is there a role for genes in exercise-induced atrial cardiomyopathy? *Heart Lung Circ* 2018;27:1093–8. 10.1016/j.hlc.2018.03.028 [PubMed: 29706494]
25. Lampert R, Chung EH, Ackerman MJ, Arroyo AR, Darden D, Deo R, et al. 2024 HRS expert consensus statement on arrhythmias in the athlete: evaluation, treatment, and return to play. *Heart Rhythm* 2024;21:e151–252. 10.1016/j.hrthm.2024.05.018 [PubMed: 38763377]
26. Khan AK, Lee HJ, Hills MT, Moss M, Lara B, Baumann R, et al. Impact of atrial fibrillation and atrial fibrillation therapies on sports performance in athletes. *Heart Rhythm* 2024;22:2562–9. 10.1016/j.hrthm.2024.11.020 [PubMed: 39557377]
27. Furlanello F, Lupo P, Pittalis M, Foresti S, Vitali-Serdoz L, Francia P, et al. Radiofrequency catheter ablation of atrial fibrillation in athletes referred for disabling symptoms preventing usual training schedule and sport competition. *J Cardiovasc Electrophysiol* 2008;19:457–62. 10.1111/j.1540-8167.2007.01077.x [PubMed: 18266680]
28. Koopman P, Nuyens D, Garweg C, La Gerche A, De Buck S, Van Casteren L, et al. Efficacy of radiofrequency catheter ablation in athletes with atrial fibrillation. *Europace* 2011;13:1386–93. 10.1093/europace/eur142 [PubMed: 21646230]
29. Mandsager KT, Phelan DM, Diab M, Baranowski B, Saliba WI, Tarakji KG, et al. Outcomes of pulmonary vein isolation in athletes. *JACC Clin Electrophysiol* 2020;6:1265–74. 10.1016/j.jacep.2020.05.009 [PubMed: 33092753]
30. Joglar JA, Chung MK, Armbruster AL, Benjamin EJ, Chyou JY, Cronin EM, et al. 2023 ACC/AHA/ACCP/HRS guideline for the diagnosis and management of atrial fibrillation: a report of the American College of Cardiology/American Heart Association joint committee on clinical practice guidelines. *Circulation* 2024;149:e1–156. 10.1161/CIR.0000000000001193 [PubMed: 38033089]
31. Van Gelder IC, Rienstra M, Bunting KV, Casado-Arroyo R, Caso V, Crijns HJGM, et al. 2024 ESC guidelines for the management of atrial fibrillation developed in collaboration with the European association for cardio-thoracic surgery (EACTS). *Eur Heart J* 2024;45:3314–414. 10.1093/eurheartj/ehae176 [PubMed: 39210723]
32. Kim JH, Baggish AL, Levine BD, Ackerman MJ, Day SM, Dineen EH, et al. Clinical considerations for competitive sports participation for athletes with cardiovascular abnormalities: a scientific statement from the American Heart Association and American College of Cardiology. *J Am Coll Cardiol* 2025;85:1059–108. 10.1016/j.jacc.2024.12.025 [PubMed: 39976316]
33. Talan DA, Bauernfeind RA, Ashley WW, Kanakis C Jr, Rosen KM. Twenty-four hour continuous ECG recordings in long-distance runners. *Chest* 1982;82:19–24. 10.1378/chest.82.1.19 [PubMed: 7083929]

34. D'Souza A, Sharma S, Boyett MR. CrossTalk opposing view: bradycardia in the trained athlete is attributable to a downregulation of a pacemaker channel in the sinus node. *J Physiol* 2015;593:1749–51. 10.1113/jphysiol.2014.284356 [PubMed: 25871551]
35. Mesirca P, Nakao S, Nissen SD, Forte G, Anderson C, Trussell T, et al. Intrinsic electrical remodeling underlies atrioventricular block in athletes. *Circ Res* 2021;129:e1–20. 10.1161/CIRCRESAHA.119.316386 [PubMed: 33849278]
36. Baldesberger S, Bauersfeld U, Candinas R, Seifert B, Zuber M, Ritter M, et al. Sinus node disease and arrhythmias in the long-term follow-up of former professional cyclists. *Eur Heart J* 2008;29:71–8. 10.1093/eurheartj/ehm555 [PubMed: 18065754]
37. Andersen K, Farahmand B, Ahlbom A, Held C, Ljunghall S, Michaëlsson K, et al. Risk of arrhythmias in 52 755 long-distance cross-country skiers: a cohort study. *Eur Heart J* 2013;34:3624–31. 10.1093/eurheartj/eh188 [PubMed: 23756332]
38. Svedberg N, Sundström J, James S, Hållmarker U, Hambraeus K, Andersen K. Long-term incidence of bradycardia and pacemaker implantations among cross-country skiers: a cohort study. *Circulation* 2024;150:1161–70. 10.1161/CIRCULATIONAHA.123.068280 [PubMed: 39101218]
39. Cavigli L, Zorzi A, Spadotto V, Gismondi A, Sisti N, Valentini F, et al. The acute effects of an ultramarathon on biventricular function and ventricular arrhythmias in master athletes. *Eur Heart J Cardiovasc Imaging* 2022;23:423–30. 10.1093/ehjci/jeab017 [PubMed: 33544827]
40. Pizzolato M, Graziano F, Bondarev S, Corrado D, Zorzi A. Incidence and characteristics of premature ventricular beats at exercise testing for preparticipation screening. *J Sports Med Phys Fitness* 2024;64:846–8. 10.23736/S0022-4707.24.16084-7 [PubMed: 38863422]
41. Zorzi A, Mastella G, Cipriani A, Berton G, Del Monte A, Gusella B, et al. Burden of ventricular arrhythmias at 12-lead 24-hour ambulatory ECG monitoring in middle-aged endurance athletes versus sedentary controls. *Eur J Prev Cardiol* 2018;25:2003–11. 10.1177/2047487318797396 [PubMed: 30160531]
42. Di Florio A, Fusi C, Anselmi F, Cavigli L, Focardi M, Cameli M, et al. Clinical management of young competitive athletes with premature ventricular beats: a prospective cohort study. *Int J Cardiol* 2021;330:59–64. 10.1016/j.ijcard.2021.02.021 [PubMed: 33582199]
43. Heidbuchel H, Arbelo E, D'Ascenzi F, Borjesson M, Boveda S, Castelletti S, et al. Recommendations for participation in leisure-time physical activity and competitive sports of patients with arrhythmias and potentially arrhythmogenic conditions. Part 2: ventricular arrhythmias, channelopathies, and implantable defibrillators. *Europace* 2021;23:147–8. 10.1093/europace/eaab106 [PubMed: 32596731]
44. Zorzi A, D'Ascenzi F, Andreini D, Castelletti S, Casella M, Cavarretta E, et al. Interpretation and management of premature ventricular beats in athletes: an expert opinion document of the Italian society of sports cardiology (SICSPORT). *Int J Cardiol* 2023;391:131220. 10.1016/j.ijcard.2023.131220 [PubMed: 37517780]
45. Cipriani A, Zorzi A, Sarto P, Donini M, Rigato I, Bariani R, et al. Predictive value of exercise testing in athletes with ventricular ectopy evaluated by cardiac magnetic resonance. *Heart Rhythm* 2019;16:239–48. 10.1016/j.hrthm.2018.08.029 [PubMed: 30172028]
46. Heidbuchel H, Prior DL, La Gerche A. Ventricular arrhythmias associated with long-term endurance sports: what is the evidence? *Br J Sports Med* 2012;46 Suppl 1:i44–50. 10.1136/bjsports-2012-091162 [PubMed: 23097479]
47. Heidbüchel H, Hoogsteen J, Fagard R, Vanhees L, Ector H, Willems R, et al. High prevalence of right ventricular involvement in endurance athletes with ventricular arrhythmias. Role of an electrophysiologic study in risk stratification. *Eur Heart J* 2003;24:1473–80. 10.1016/S0195-668X(03)00282-3 [PubMed: 12919770]
48. D'Ambrosio P, De Paepe J, Janssens K, Mitchell AM, Rowe SJ, Spencer LW, et al. Arrhythmias and structural remodeling in lifelong and retired master endurance athletes. *J Sport Health Sci* 2025;14:101043. 10.1016/j.jsbs.2025.101043 [PubMed: 40273982]
49. Corrado D, Drezner JA, D'Ascenzi F, Zorzi A. How to evaluate premature ventricular beats in the athlete: critical review and proposal of a diagnostic algorithm. *Br J Sports Med* 2020;54:1142–8. 10.1136/bjsports-2018-100529 [PubMed: 31481389]

50. Aengevaeren VL, Mosterd A, Sharma S, Prakken NHJ, Möhlenkamp S, Thompson PD, et al. Exercise and coronary atherosclerosis: observations, explanations, relevance, and clinical management. *Circulation* 2020;141:1338–50. 10.1161/CIRCULATIONAHA.119.044467 [PubMed: 32310695]
51. Möhlenkamp S, Lehmann N, Breuckmann F, Bröcker-Preuss M, Nassenstein K, Halle M, et al. Running: the risk of coronary events : prevalence and prognostic relevance of coronary atherosclerosis in marathon runners. *Eur Heart J* 2008;29:1903–10. 10.1093/eurheartj/ehn163 [PubMed: 18426850]
52. DeFina LF, Radford NB, Barlow CE, Willis BL, Leonard D, Haskell WL, et al. Association of all-cause and cardiovascular mortality with high levels of physical activity and concurrent coronary artery calcification. *JAMA Cardiol* 2019;4:174–81. 10.1001/jamacardio.2018.4628 [PubMed: 30698608]
53. Papatheodorou E, Aengevaeren VL, Eijsvogels TMH, AlFakih K, Hughes RK, Merghani A, et al. Prevalence of coronary atherosclerosis in female masters endurance athletes. *Circulation* 2024;150:1478–80. 10.1161/CIRCULATIONAHA.124.069484 [PubMed: 39466884]
54. Claessen G, Eijsvogels TMH, Albert CM, Baggish AL, Levine BD, Marijon E, et al. Coronary atherosclerosis in athletes: emerging concepts and preventive strategies. *Eur Heart J* 2025;46:890–903. 10.1093/eurheartj/ehae927 [PubMed: 39791533]
55. Puri R, Nicholls SJ, Shao M, Kataoka Y, Uno K, Kapadia SR, et al. Impact of statins on serial coronary calcification during atheroma progression and regression. *J Am Coll Cardiol* 2015;65:1273–82. 10.1016/j.jacc.2015.01.036 [PubMed: 25835438]
56. Aengevaeren VL, Claessen G, Eijsvogels TM. Coronary atherosclerosis in athletes: recent insights and clinical considerations. *Br J Sports Med* 2024;58:574–6. 10.1136/bjsports-2023-107938 [PubMed: 38503469]
57. Vine M, Daucourt C, Besson C, Neyroud D, Kayser B, Meyer P, et al. Cardiovascular risk factors among endurance masters athletes from Switzerland. *Eur J Prev Cardiol* 2024; 10.1093/eurjpc/zwae287
58. Berge K, Janssen SLJE, Velthuis BK, Myhre PL, Mosterd A, Omland T, et al. Predictors of coronary atherosclerosis in middle-aged and older athletes: the MARC-2 study. *Eur Heart J Cardiovasc Imaging* 2025;26:461–470. 10.1093/ehjci/jeae317 [PubMed: 39657626]
59. Zouhal H, Jacob C, Delamarche P, Gratas-Delamarche A. Catecholamines and the effects of exercise, training and gender. *Sports Med* 2008;38:401–23. 10.2165/00007256-200838050-00004 [PubMed: 18416594]
60. Aengevaeren VL, Mosterd A, Bakker EA, Braber TL, Nathoe HM, Sharma S, et al. Exercise volume versus intensity and the progression of coronary atherosclerosis in middle-aged and older athletes: findings from the MARC-2 study. *Circulation* 2023;147:993–1003. 10.1161/CIRCULATIONAHA.122.061173 [PubMed: 36597865]
61. Bouassida A, Latiri I, Bouassida S, Zalleg D, Zaouali M, Feki Y, et al. Parathyroid hormone and physical exercise: a brief review. *J Sports Sci Med* 2006;05:367–74. <https://www.jssm.org/jssm-05-367.xml-Fulltext#>
62. Kohrt WM, Wherry SJ, Wolfe P, Sherk VD, Wellington T, Swanson CM, et al. Maintenance of Serum ionized calcium during exercise attenuates parathyroid hormone and bone resorption responses. *J Bone Miner Res* 2018;33:1326–34. 10.1002/jbmr.3428 [PubMed: 29572961]
63. Radford NB, DeFina LF, Leonard D, Barlow CE, Willis BL, Gibbons LW, et al. Cardiorespiratory fitness, coronary artery calcium, and cardiovascular disease events in a cohort of generally healthy middle-age men: results from the cooper center longitudinal study. *Circulation* 2018;137:1888–95. 10.1161/CIRCULATIONAHA.117.032708 [PubMed: 29343464]
64. Amson Y, Rozanski A, Gransar H, Hayes SW, Friedman JD, Thomson LEJ, et al. Impact of exercise on the relationship between CAC scores and allcause mortality. *JACC Cardiovasc Imaging* 2017;10:1461–8. 10.1016/j.jcmg.2016.12.030 [PubMed: 28528154]
65. Visseren FLJ, Mach F, Smulders YM, Carballo D, Koskinas KC, Bäck M, et al. 2021 ESC guidelines on cardiovascular disease prevention in clinical practice. *Eur Heart J* 2021;42:3227–337. 10.1093/eurheartj/ehab484 [PubMed: 34458905]

66. Budoff MJ, Kinninger A, Gransar H, Achenbach S, Al-Mallah M, Bax JJ, et al. When does a calcium score equate to secondary prevention?: insights from the multinational CONFIRM registry. *JACC Cardiovasc Imaging* 2023;16:1181–9. 10.1016/j.jcmg.2023.03.008 [PubMed: 37227328]
67. Peng AW, Dardari ZA, Blumenthal RS, Dzaye O, Obisesan OH, Iftekhhar Uddin SM, et al. Very high coronary artery calcium (≥ 1000) and association with cardiovascular disease events, non-cardiovascular disease outcomes, and mortality: results from MESA. *Circulation* 2021;143:1571–83. 10.1161/CIRCULATIONAHA.120.050545 [PubMed: 33650435]
68. McClelland RL, Jorgensen NW, Budoff M, Blaha MJ, Post WS, Kronmal RA, et al. 10-Year coronary heart disease risk prediction using coronary artery calcium and traditional risk factors: derivation in the MESA (multi-ethnic study of atherosclerosis) with validation in the HNR (Heinz Nixdorf recall) study and the DHS (Dallas heart study). *J Am Coll Cardiol* 2015;66:1643–53. 10.1016/j.jacc.2015.08.035 [PubMed: 26449133]
69. Khera A, Budoff MJ, O'Donnell CJ, Ayers CA, Locke J, de Lemos JA, et al. Astronaut cardiovascular health and risk modification (astro-CHARM) coronary calcium atherosclerotic cardiovascular disease risk calculator. *Circulation* 2018;138:1819–27. 10.1161/CIRCULATIONAHA.118.033505 [PubMed: 30354651]
70. Kim JH, Malhotra R, Chiampas G, d'Hemecourt P, Troyanos C, Cianca J, et al. Cardiac arrest during long-distance running races. *N Engl J Med* 2012;366:130–40. 10.1056/NEJMoa1106468 [PubMed: 22236223]
71. Harris KM, Creswell LL, Haas TS, Thomas T, Tung M, Isaacson E, et al. Death and cardiac arrest in U.S. Triathlon participants, 1985 to 2016: a case series. *Ann Intern Med* 2017;167:529–35. 10.7326/M17-0847 [PubMed: 28975231]
72. Marijon E, Uy-Evanado A, Reinier K, Teodorescu C, Narayanan K, Jouven X, et al. Sudden cardiac arrest during sports activity in middle age. *Circulation* 2015;131:1384–91. 10.1161/CIRCULATIONAHA.114.011988 [PubMed: 25847988]
73. Maron DJ, Hochman JS, Reynolds HR, Bangalore S, O'Brien SM, Boden WE, et al. Initial invasive or conservative strategy for stable coronary disease. *N Engl J Med* 2020;382:1395–407. 10.1056/NEJMoa1915922 [PubMed: 32227755]
74. Vrints C, Andreotti F, Koskinas KC, Rossello X, Adamo M, Ainslie J, et al. 2024 ESC guidelines for the management of chronic coronary syndromes. *Eur Heart J* 2024;45:3415–537. 10.1093/eurheartj/ehae177 [PubMed: 39210710]
75. Iskandar A, Thompson PD. A meta-analysis of aortic root size in elite athletes. *Circulation* 2013;127:791–8. 10.1161/CIRCULATIONAHA.112.000974 [PubMed: 23322885]
76. Pelliccia A, Di Paolo FM, De Blasiis E, Quattrini FM, Pisicchio C, Guerra E, et al. Prevalence and clinical significance of aortic root dilation in highly trained competitive athletes. *Circulation* 2010;122:698–706; 693 p following 706. 10.1161/CIRCULATIONAHA.109.901074 [PubMed: 20679553]
77. Gati S, Malhotra A, Sedgwick C, Papamichael N, Dhutia H, Sharma R, et al. Prevalence and progression of aortic root dilatation in highly trained young athletes. *Heart* 2019;105:920–5. 10.1136/heartjnl-2018-314288 [PubMed: 30910821]
78. Boraita A, Heras ME, Morales F, Marina-Breyse M, Canda A, Rabadan M, et al. Reference values of aortic root in male and female white elite athletes according to sport. *Circ Cardiovasc Imaging* 2016;9:e005292. 10.1161/CIRCIMAGING.116.005292 [PubMed: 27729365]
79. Kinoshita N, Mimura J, Obayashi C, Katsukawa F, Onishi S, Yamazaki H. Aortic root dilatation among young competitive athletes: echocardiographic screening of 1929 athletes between 15 and 34 years of age. *Am Heart J* 2000;139:723–8. 10.1016/S0002-8703(00)90055-3 [PubMed: 10740158]
80. Gentry JLC, Joshi PH, Maroules CD, Ayers CR, De Lemos JA, Aagaard P, et al. Ascending aortic dimensions in former national football league athletes. *Circ Cardiovasc Imaging* 2017;10:e006852. 10.1161/CIRCIMAGING.117.006852 [PubMed: 29122845]
81. Kay S, Moore BM, Moore L, Seco M, Barnes C, Marshman D, et al. Rugby player's aorta: alarming prevalence of ascending aortic dilatation and effacement in elite rugby players. *Heart Lung Circ* 2020;29:196–201. 10.1016/j.hlc.2019.06.714 [PubMed: 31494040]

82. Tso JV, Turner CG, Liu C, Miller AB, Eapen DJ, Sperling LS, et al. Exercise blood pressure changes and aortic dilatation in male masters endurance athletes. *Eur J Prev Cardiol* 2023;30:e18–20. 10.1093/eurjpc/zwac250 [PubMed: 36306424]
83. Boraita A, Morales-Acuna F, Marina-Breyse M, Heras ME, Canda A, Fuentes ME, et al. Bicuspid aortic valve behaviour in elite athletes. *Eur Heart J Cardiovasc Imaging* 2019;20:772–80. 10.1093/ehjci/jez001 [PubMed: 30715268]
84. Braverman AC. Aortic involvement in patients with a bicuspid aortic valve. *Heart* 2011;97:506–13. 10.1136/hrt.2009.183871 [PubMed: 21339321]
85. Isselbacher EM. Thoracic and abdominal aortic aneurysms. *Circulation* 2005;111:816–28. 10.1161/01.CIR.0000154569.08857.7A [PubMed: 15710776]
86. Turkbey EB, Jain A, Johnson C, Redheuil A, Arai AE, Gomes AS, et al. Determinants and normal values of ascending aortic diameter by age, gender, and race/ethnicity in the multi-ethnic study of atherosclerosis (MESA). *J Magn Reson Imaging* 2014;39:360–8. 10.1002/jmri.24183 [PubMed: 23681649]
87. Schreurs BA, Hopman MTE, Bakker CM, Duijnhouwer AL, van Royen N, Thompson PD, et al. Associations of lifelong exercise characteristics with valvular function and aortic diameters in patients with a bicuspid aortic valve. *J Am Heart Assoc* 2024;13:e031850. 10.1161/JAHA.123.031850 [PubMed: 38293944]
88. D’Ascenzi F, Cavigli L, Cameli M, Claessen G, van Craenenbroeck EM, Cavarretta E, et al. Sport PRactice and its effects on aortic size and valve function in bicuspid aortic valve disease: a cross-sectional report from the SPREAD study. *Br J Sports Med* 2024;58:1419–25. 10.1136/bjsports-2023-107772 [PubMed: 39153748]
89. MacDougall JD, Tuxen D, Sale DG, Moroz JR, Sutton JR. Arterial blood pressure response to heavy resistance exercise. *J Appl Physiol* 1985;58:785–90. 10.1152/jappl.1985.58.3.785 [PubMed: 3980383]
90. Lai AF, Braverman AC. Endurance exercise following ascending thoracic aortic aneurysm resection in bicuspid aortic valve aortopathy. *JAMA Cardiol* 2022;7:772–3. 10.1001/jamacardio.2022.1410 [PubMed: 35675088]
91. Isselbacher EM, Preventza O, Hamilton Black J 3rd, Augoustides JG, Beck AW, Bolen MA, et al. 2022 ACC/AHA guideline for the diagnosis and management of aortic disease: a report of the American Heart Association/American College of Cardiology joint committee on clinical practice guidelines. *Circulation* 2022;146:e334–482. 10.1161/CIR.0000000000001106 [PubMed: 36322642]
92. Baggish AL, Battle RW, Beaver TA, Border WL, Douglas PS, Kramer CM, et al. Recommendations on the use of multimodality cardiovascular imaging in young adult competitive athletes: a report from the American society of echocardiography in collaboration with the society of cardiovascular computed tomography and the society for cardiovascular magnetic resonance. *J Am Soc Echocardiogr* 2020;33:523–49. 10.1016/j.echo.2020.02.009 [PubMed: 32362332]
93. Pelliccia A, Caselli S, Sharma S, Basso C, Bax JJ, Corrado D, et al. European association of preventive cardiology (EAPC) and European association of cardiovascular imaging (EACVI) joint position statement: recommendations for the indication and interpretation of cardiovascular imaging in the evaluation of the athlete’s heart. *Eur Heart J* 2018;39:1949–69. 10.1093/eurheartj/ehx532 [PubMed: 29029207]
94. Davies RR, Gallo A, Coady MA, Tellides G, Botta DM, Burke B, et al. Novel measurement of relative aortic size predicts rupture of thoracic aortic aneurysms. *Ann Thorac Surg* 2006;81:169–77. 10.1016/j.athoracsurg.2005.06.026 [PubMed: 16368358]
95. Saura D, Dulgheru R, Caballero L, Bernard A, Kou S, Gonjilashvili N, et al. Two-dimensional transthoracic echocardiographic normal reference ranges for proximal aorta dimensions: results from the EACVI NORRE study. *Eur Heart J Cardiovasc Imaging* 2017;18:167–79. 10.1093/ehjci/jew053 [PubMed: 27044913]
96. Zafar MA, Li Y, Rizzo JA, Charilaou P, Saeyeldin A, Velasquez CA, et al. Height alone, rather than body surface area, suffices for risk estimation in ascending aortic aneurysm. *J Thorac Cardiovasc Surg* 2018;155:1938–50. 10.1016/j.jtcvs.2017.10.140 [PubMed: 29395211]
97. Masri A, Kalahasti V, Svensson LG, Alashi A, Schoenhagen P, Roselli EE, et al. Aortic cross-sectional area/height ratio and outcomes in patients with bicuspid aortic

valve and a dilated ascending aorta. *Circ Cardiovasc Imaging* 2017;10:e006249. 10.1161/CIRCIMAGING.116.006249 [PubMed: 28592593]

98. Masri A, Kalahasti V, Svensson LG, Roselli EE, Johnston D, Hammer D, et al. Aortic cross-sectional area/height ratio and outcomes in patients with a trileaflet aortic valve and a dilated aorta. *Circulation* 2016;134:1724–37. 10.1161/CIRCULATIONAHA.116.022995 [PubMed: 27770001]
99. Svensson LG, Kim KH, Lytle BW, Cosgrove DM. Relationship of aortic cross-sectional area to height ratio and the risk of aortic dissection in patients with bicuspid aortic valves. *J Thorac Cardiovasc Surg* 2003;126:892–3. 10.1016/S0022-5223(03)00608-1 [PubMed: 14502185]
100. Wu J, Zafar MA, Li Y, Saeyeldin A, Huang Y, Zhao R, et al. Ascending aortic length and risk of aortic adverse events: the neglected dimension. *J Am Coll Cardiol* 2019;74:1883–94. 10.1016/j.jacc.2019.07.078 [PubMed: 31526537]
101. Devereux RB, de Simone G, Arnett DK, Best LG, Boerwinkle E, Howard BV, et al. Normal limits in relation to age, body size and gender of two-dimensional echocardiographic aortic root dimensions in persons ≥ 15 years of age. *Am J Cardiol* 2012;110:1189–94. 10.1016/j.amjcard.2012.05.063 [PubMed: 22770936]
102. Campens L, Demulier L, De Groote K, Vandekerckhove K, De Wolf D, Roman MJ, et al. Reference values for echocardiographic assessment of the diameter of the aortic root and ascending aorta spanning all age categories. *Am J Cardiol* 2014;114:914–20. 10.1016/j.amjcard.2014.06.024 [PubMed: 25092193]
103. Muntner P, Shimbo D, Carey RM, Charleston JB, Gaillard T, Misra S, et al. Measurement of blood pressure in humans: a scientific statement from the American Heart Association. *Hypertension* 2019;73:e35–66. 10.1161/HYP.0000000000000087 [PubMed: 30827125]
104. Erbel R, Aboyans V, Boileau C, Bossone E, Bartolomeo RD, Eggebrecht H, et al. 2014 ESC guidelines on the diagnosis and treatment of aortic diseases: document covering acute and chronic aortic diseases of the thoracic and abdominal aorta of the adult. The task force for the diagnosis and treatment of aortic diseases of the European Society of Cardiology (ESC). *Eur Heart J* 2014;35:2873–926. 10.1093/eurheartj/ehu281 [PubMed: 25173340]
105. Kim JH, Dickert NW. Athletes with cardiovascular disease and competitive sports eligibility: progress and challenges ahead. *JAMA Cardiol* 2022;7:663–4. 10.1001/jamacardio.2022.0806 [PubMed: 35544077]
106. Javed W, Malhotra A, Swoboda P. Cardiac magnetic resonance assessment of athletic myocardial fibrosis; benign bystander or malignant marker? *Int J Cardiol* 2024;394:131382. 10.1016/j.ijcard.2023.131382 [PubMed: 37741350]
107. Sánchez-Somonte P, Quinto L, Garre P, Zaraket F, Alarcón F, Borràs R, et al. Scar channels in cardiac magnetic resonance to predict appropriate therapies in primary prevention. *Heart Rhythm* 2021;18:1336–43. 10.1016/j.hrthm.2021.04.017 [PubMed: 33892202]
108. Ragab H, Lund GK, Breitsprecher L, Sinn MR, Muellerleile K, Cavus E, et al. Prevalence and pattern of focal and potential diffuse myocardial fibrosis in male and female marathon runners using contrast-enhanced cardiac magnetic resonance. *Eur Radiol* 2023;33:4648–56. 10.1007/s00330-023-09416-3 [PubMed: 36683089]
109. Farooq M, Brown LAE, Fitzpatrick A, Broadbent DA, Wahab A, Klassen JRL, et al. Identification of non-ischaemic fibrosis in male veteran endurance athletes, mechanisms and association with premature ventricular beats. *Sci Rep* 2023;13:14640. 10.1038/s41598-023-40252-z [PubMed: 37669972]
110. Domenech-Ximenes B, Sanz-de la Garza M, Prat-González S, Sepúlveda-Martínez A, Crispí F, Duran-Fernandez K, et al. Prevalence and pattern of cardiovascular magnetic resonance late gadolinium enhancement in highly trained endurance athletes. *J Cardiovasc Magn Reson* 2020;22:62. 10.1186/s12968-020-00660-w [PubMed: 32878630]
111. Tahir E, Starekova J, Muellerleile K, von Stritzky A, Münch J, Avanesov M, et al. Myocardial fibrosis in competitive triathletes detected by contrast-enhanced CMR correlates with exercise-induced hypertension and competition history. *JACC Cardiovasc Imaging* 2018;11:1260–70. 10.1016/j.jcmg.2017.09.016 [PubMed: 29248656]
112. Wilson M, O'Hanlon R, Prasad S, Deighan A, Macmillan P, Oxborough D, et al. Diverse patterns of myocardial fibrosis in lifelong, veteran endurance athletes. *J Appl Physiol* 2011;110:1622–6. 10.1152/japplphysiol.01280.2010 [PubMed: 21330616]

113. Breuckmann F, Möhlenkamp S, Nassenstein K, Lehmann N, Ladd S, Schmermund A, et al. Myocardial late gadolinium enhancement: prevalence, pattern, and prognostic relevance in marathon runners. *Radiology* 2009;251:50–7. 10.1148/radiol.2511081118 [PubMed: 19332846]
114. Andresen K, Klæboe LG, Lie ØH, Broch K, Kvaslerud AB, Bosse G, et al. No adverse association between exercise exposure and diffuse myocardial fibrosis in male endurance athletes. *Sci Rep* 2024;14:6581. 10.1038/s41598-024-57233-5 [PubMed: 38503845]
115. Missenard O, Gabaudan C, Astier H, Desmots F, Garnotel E, Massoure PL. Absence of cardiac damage induced by long-term intensive endurance exercise training: a cardiac magnetic resonance and exercise echocardiography analysis in masters athletes. *Am J Prev Cardiol* 2021;7:100196. 10.1016/j.ajpc.2021.100196 [PubMed: 34611636]
116. Małek ŁA, Barczuk-Falęcka M, Werys K, Czajkowska A, Mróz A, Witek K, et al. Cardiovascular magnetic resonance with parametric mapping in long-term ultra-marathon runners. *Eur J Radiol* 2019;117:89–94. 10.1016/j.ejrad.2019.06.001 [PubMed: 31307657]
117. Pujadas S, Doñate M, Li CH, Merchan S, Cabanillas A, Alomar X, et al. Myocardial remodelling and tissue characterisation by cardiovascular magnetic resonance (CMR) in endurance athletes. *BMJ Open Sport Exerc Med* 2018;4:e000422. 10.1136/bmjsem-2018-000422
118. Abdullah SM, Barkley KW, Bhella PS, Hastings JL, Matulevicius S, Fujimoto N, et al. Lifelong physical activity regardless of dose is not associated with myocardial fibrosis. *Circ Cardiovasc Imaging* 2016;9:e005511. 10.1161/CIRCIMAGING.116.005511 [PubMed: 27903541]
119. Bohm P, Schneider G, Linneweber L, Rentzsch A, Krämer N, Abdul-Khaliq H, et al. Right and left ventricular function and mass in male elite master athletes: a controlled contrast-enhanced cardiovascular magnetic resonance study. *Circulation* 2016;133:1927–35. 10.1161/CIRCULATIONAHA.115.020975 [PubMed: 27073129]
120. Arbelo E, Protonotarios A, Gimeno JR, Arbustini E, Barriales-Villa R, Basso C, et al. 2023 ESC guidelines for the management of cardiomyopathies. *Eur Heart J* 2023;44:3503–626. 10.1093/eurheartj/ehad194 [PubMed: 37622657]
121. Grigoratos C, Pantano A, Meschisi M, Gaeta R, Ait-Ali L, Barison A, et al. Clinical importance of late gadolinium enhancement at right ventricular insertion points in otherwise normal hearts. *Int J Cardiovasc Imaging* 2020;36:913–20. 10.1007/s10554-020-01783-y [PubMed: 32026265]
122. Swift AJ, Rajaram S, Capener D, Elliot C, Condliffe R, Wild JM, et al. LGE patterns in pulmonary hypertension do not impact overall mortality. *JACC Cardiovasc Imaging* 2014;7:1209–17. 10.1016/j.jcmg.2014.08.014 [PubMed: 25496540]
123. Sawant AC, Bhonsale A, te Riele AS, Tichnell C, Murray B, Russell SD, et al. Exercise has a disproportionate role in the pathogenesis of arrhythmogenic right ventricular dysplasia/cardiomyopathy in patients without desmosomal mutations. *J Am Heart Assoc* 2014;3:e001471. 10.1161/JAHA.114.001471 [PubMed: 25516436]
124. La Gerche A Exercise-induced arrhythmogenic (right ventricular) cardiomyopathy is real...if you consider it. *JACC Cardiovasc Imaging* 2021;14:159–61. 10.1016/j.jcmg.2020.09.014 [PubMed: 33221208]
125. Venlet J, Piers SR, Jongbloed JD, Androulakis AF, Naruse Y, den Uijl DW, et al. Isolated subepicardial right ventricular outflow tract scar in athletes with ventricular tachycardia. *J Am Coll Cardiol* 2017;69:497–507. 10.1016/j.jacc.2016.11.041 [PubMed: 28153106]
126. Zorzi A, Perazzolo Marra M, Rigato I, De Lazzari M, Susana A, Niero A, et al. Nonischemic left ventricular scar as a substrate of life-threatening ventricular arrhythmias and sudden cardiac death in competitive athletes. *Circ Arrhythm Electrophysiol* 2016;9:e004229. 10.1161/CIRCEP.116.004229 [PubMed: 27390211]
127. La Gerche A, Robberecht C, Kuiperi C, Nuyens D, Willems R, de Rave IT, et al. Lower than expected desmosomal gene mutation prevalence in endurance athletes with complex ventricular arrhythmias of right ventricular origin. *Heart* 2010;96:1268–74. 10.1136/hrt.2009.189621 [PubMed: 20525856]
128. La Gerche A, Burns AT, Mooney DJ, Inder WJ, Taylor AJ, Bogaert J, et al. Exercise-induced right ventricular dysfunction and structural remodelling in endurance athletes. *Eur Heart J* 2012;33:998–1006. 10.1093/eurheartj/ehr397 [PubMed: 22160404]

129. Wasfy MM, Baggish AL. Endurance exercise and the right ventricle: weak link, innocent bystander, or key ingredient? *Circulation* 2016;133:1913–5. 10.1161/CIRCULATIONAHA.116.022418 [PubMed: 27073130]
130. Claessen G, De Bosscher R, Janssens K, Young P, Dausin C, Claeys M, et al. Reduced ejection fraction in elite endurance athletes: clinical and genetic overlap with dilated cardiomyopathy. *Circulation* 2024;149:1405–15. 10.1161/CIRCULATIONAHA.122.063777 [PubMed: 38109351]
131. Favere K, Van Hecke M, Eens S, Bosman M, Delputte PL, De Sutter J, et al. The influence of endurance exercise training on myocardial fibrosis and arrhythmogenesis in a coxsackievirus B3 myocarditis mouse model. *Sci Rep* 2024;14:12653. 10.1038/s41598-024-61874-x [PubMed: 38825590]
132. Haugaa KH, Rootwelt-Norberg C. Cardiac desmosomal reserve: another piece of the exercise-induced arrhythmogenic cardiomyopathy puzzle? *Eur Heart J* 2022;43:1265–7. 10.1093/eurheartj/ehab873 [PubMed: 34972201]
133. James CA, Bhonsale A, Tichnell C, Murray B, Russell SD, Tandri H, et al. Exercise increases age-related penetrance and arrhythmic risk in arrhythmogenic right ventricular dysplasia/cardiomyopathy-associated desmosomal mutation carriers. *J Am Coll Cardiol* 2013;62:1290–7. 10.1016/j.jacc.2013.06.033 [PubMed: 23871885]
134. Lie ØH, Klaboe LG, Dejgaard LA, Skjølsvik ET, Grimsø J, Bosse G, et al. Cardiac phenotypes and markers of adverse outcome in elite athletes with ventricular arrhythmias. *JACC Cardiovasc Imaging* 2021;14:148–58. 10.1016/j.jcmg.2020.07.039 [PubMed: 33129723]
135. Zaidi A, Sheikh N, Jongman JK, Gati S, Panoulas VF, Carr-White G, et al. Clinical differentiation between physiological remodeling and arrhythmogenic right ventricular cardiomyopathy in athletes with marked electrocardiographic repolarization anomalies. *J Am Coll Cardiol* 2015;65:2702–11. 10.1016/j.jacc.2015.04.035 [PubMed: 26112193]
136. La Gerche A, Claessen G, Dymarkowski S, Voigt JU, De Buck F, Vanhees L, et al. Exercise-induced right ventricular dysfunction is associated with ventricular arrhythmias in endurance athletes. *Eur Heart J* 2015;36:1998–2010. 10.1093/eurheartj/ehv202 [PubMed: 26038590]
137. Tfelt-Hansen J, Winkel BG, de Riva M, Zeppenfeld K. 2022 ESC guidelines for the management of patients with ventricular arrhythmias and the prevention of sudden cardiac death. *Eur Heart J* 2022;43:3997–4126. 10.1093/eurheartj/ehac262 [PubMed: 36017572]
138. Darden D, Scheinman MM, Hoffmayer KS. Exercise-induced arrhythmogenic right ventricular cardiomyopathy: reverse remodeling with detraining. *HeartRhythm Case Rep* 2022;8:599–603. 10.1016/j.hrcr.2022.06.003 [PubMed: 36147714]
139. Clark J, Bakhit Y, Motwani M. Exercise-induced right ventricular cardiomyopathy in an endurance cyclist: a case report. *Eur Heart J Case Rep* 2023;7:ytad002. 10.1093/ehjcr/ytad002 [PubMed: 36694872]
140. McDonagh TA, Metra M, Adamo M, Gardner RS, Baumbach A, Böhm M, et al. 2021 ESC guidelines for the diagnosis and treatment of acute and chronic heart failure. *Eur Heart J* 2021;42:3599–726. 10.1093/eurheartj/ehab368 [PubMed: 34447992]
141. Gerber Y, Gabriel KP, Jacobs DR Jr, Liu JY, Rana JS, Sternfeld B, et al. The relationship of cardiorespiratory fitness, physical activity, and coronary artery calcification to cardiovascular disease events in CARDIA participants. *Eur J Prev Cardiol* 2025;32:52–62. 10.1093/eurjpc/zwae272 [PubMed: 39158112]
142. Eijsvogels TM, Fernandez AB, Thompson PD. Are there deleterious cardiac effects of acute and chronic endurance exercise? *Physiol Rev* 2016;96:99–125. 10.1152/physrev.00029.2014 [PubMed: 26607287]
143. Baggish AL, Weiner RB, Kanayama G, Hudson JI, Lu MT, Hoffmann U, et al. Cardiovascular toxicity of illicit anabolic-androgenic steroid use. *Circulation* 2017;135:1991–2002. 10.1161/CIRCULATIONAHA.116.026945 [PubMed: 28533317]
144. Krishnan S, Guseh JS, Chukumerije M, Grant AJ, Dean PN, Hsu JJ, et al. Racial disparities in sports cardiology: a review. *JAMA Cardiol* 2024;9:935. 10.1001/jamacardio.2024.1899 [PubMed: 39018059]
145. Corrado D, van Tintelen PJ, McKenna WJ, Hauer RNW, Anastakis A, Asimaki A, et al. Arrhythmogenic right ventricular cardiomyopathy: evaluation of the current diagnostic criteria

and differential diagnosis. *Eur Heart J* 2020;41:1414–29. 10.1093/eurheartj/ehz669 [PubMed: 31637441]

Author Manuscript

Author Manuscript

Author Manuscript

Author Manuscript

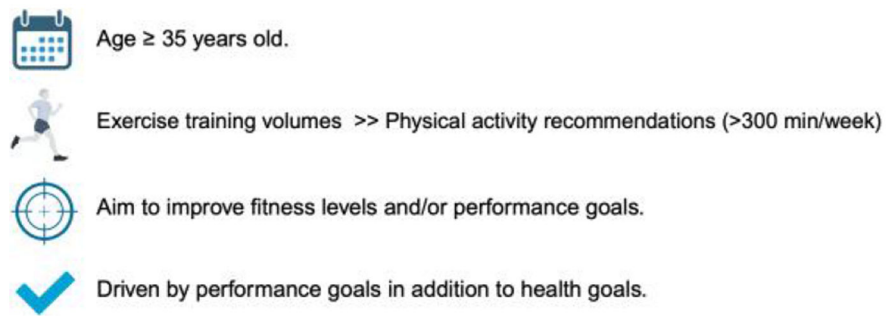


Figure 1.

Masters athletes are defined as individuals aged 35 years or older with exercise habits that consistently exceed physical activity recommendations, aimed to improve fitness and achievement of performance goals rather than health goals

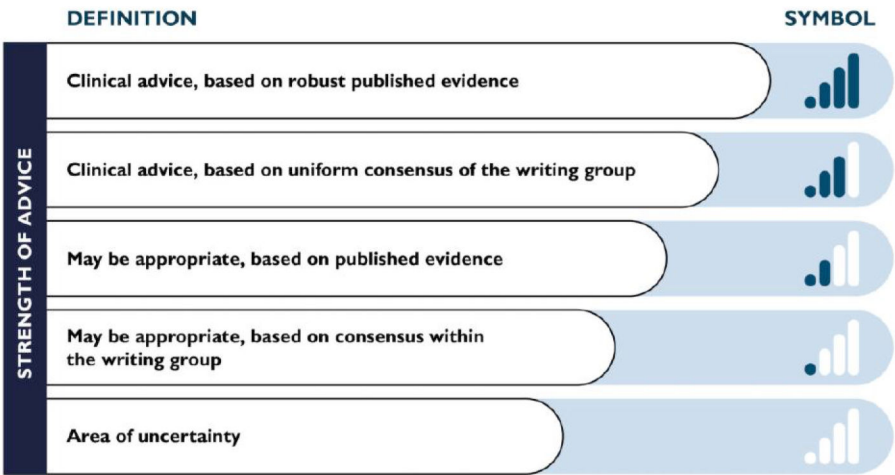


Figure 2. Explanation of ‘Strength of advice’ categories for Clinical Consensus Statements as per ESC Scientific Document Policy. Clinical advice: where there is evidence or general agreement that a given measure is clinically useful and appropriate, or evidence and general agreement that a given measure is harmful and not appropriate. May be appropriate: where there is evidence or general agreement that a given measure may be clinically useful and appropriate. Areas of uncertainty: no data available, solely based on expert opinion

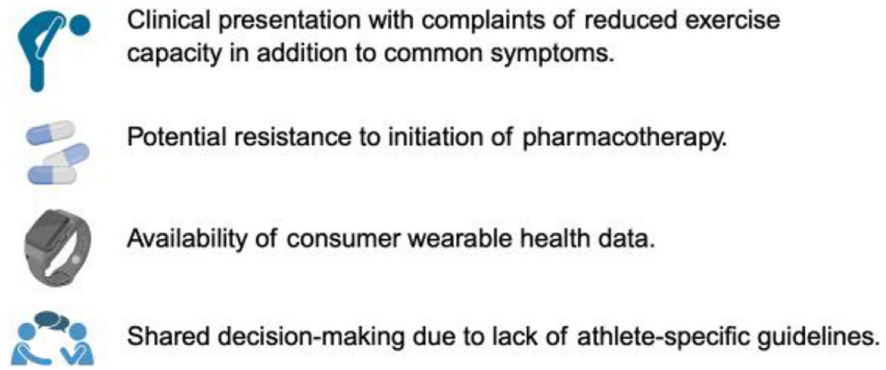


Figure 3.

The clinical assessment of Masters athletes includes a number of unique challenges, such as (i) a different clinical presentation, (ii) resistance to pharmacotherapy, (iii) availability of wearable data, and (iv) a shared decision-making approach

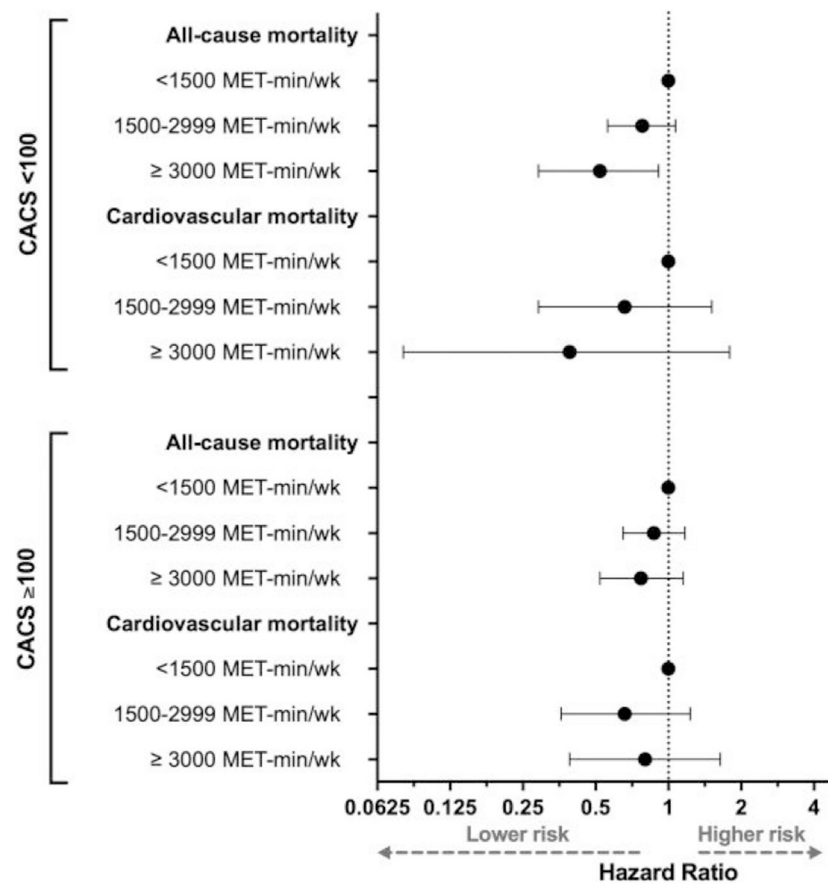


Figure 4.

All-cause and cardiovascular mortality risks in male participants of the Cooper Center Longitudinal Study ($n = 21\,758$) with distinct physical activity patterns and stratified for a CACS <100 AU vs ≥100 AU. Data were previously published,⁵² presented as hazard ratios with 95% confidence intervals, and indicate that weekly exercise volume does not attenuate event rate in men with CACS ≥100 AU

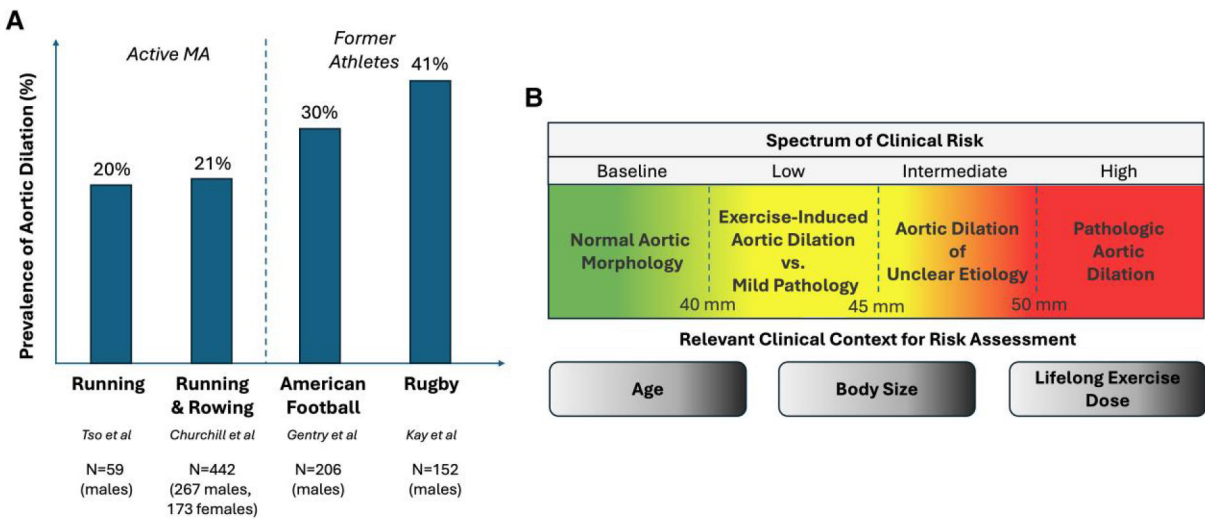


Figure 5. Prevalence of aortic dilatation (≥ 40 mm) in active Masters Athletes (MA)^{15,82} and former elite strength athletes^{80,81} (Panel A) and the spectrum of clinical risk among athletes with aortic dilatation (Panel B)

Table 1

Clinical consensus statements for Masters athletes with atrial fibrillation

#	Consideration	Strength of the advice
1	Endurance sports-related cardiovascular changes may promote AF at higher training loads, but moderate intensity physical activity should be encouraged given the many physiological and psychological health benefits.	
2	Masters athletes presenting with AF should undergo evaluation for and treatment of standard risk factors (hypertension, obstructive sleep apnea, weight control, alcohol use, and hyperthyroidism).	
3	Anticoagulation for Masters athletes should follow standard guidelines, including the use of anticoagulants for Masters athletes with a CHA ₂ DS ₂ -VA score ≥2.	
4	Among Masters athletes that meet criteria for anticoagulation, the type of sport and risk of collision/trauma should be integrated with SDM regarding future sports participation.	
5	Medical management should aim for safe sports participation, including adequate rate control during exercise and/or avoidance of exercise for two half-lives after pill-in-the-pocket treatment with Class 1C antiarrhythmics.	
6	Rhythm control is generally preferred over rate control in Masters athletes with AF to optimize performance and enhance quality of life.	
7	With SDM, pulmonary vein isolation is reasonable as an early treatment option in athletes, taking possible complications and the unknown impact on exercise tolerance into account.	
8	It is unknown whether prescribed reduction of endurance sports activity leads to lower rates of AF recurrence.	
9	Left atrial appendage occlusion devices should not be implanted for the sole purpose to avoid anticoagulation in Masters athletes with AF.	

Author Manuscript

Author Manuscript

Author Manuscript

Author Manuscript

Table 2

Clinical consensus statements for Masters athletes with bradyarrhythmias

#	Consideration	Strength of the advice
1	The clinical evaluation of Masters athletes presenting with symptoms, that may be attributable to bradyarrhythmias (i.e. dizziness, presyncope, intermittent exercise intolerance, etc.), should include a careful history, exercise testing and ambulatory rhythm monitoring to evaluate chronotropic response to exercise.	
2	In asymptomatic Masters athletes with sinus bradycardia, first degree AV block, or Mobitz 1 AV block, no intervention is required. In contrast in Masters athletes with pathological bradyarrhythmias (type second degree and third degree heart block) decision making regarding pacemaker placement should follow standard clinical guidelines.	
3	It is unknown whether prescribed reductions in sports activity among Masters athletes with chronotropic incompetence reduces the necessity for future pacemaker implantation.	
4	Among Masters athletes with subjectively or objectively reduced exercise capacity and chronotropic incompetence, it is unknown whether pacemaker implantation with rate-response functionality improves exercise capacity.	

Table 3
Clinical consensus statements for Masters athletes with ventricular arrhythmias

#	Consideration	Strength of the advice
1	In asymptomatic Masters athletes with isolated rare (1 on a resting ECG) PVCs of typical morphology (i.e. originating from the outflow tract or fascicles), no further clinical work-up is required.	
2	For Masters athletes with > 1 typical PVC, at least 1 atypical PVC on resting ECG, or symptomatic PVCs, the clinical work-up should include family and personal history, echocardiography, exercise testing, ambulatory rhythm monitoring (including a training session) and individualized longitudinal surveillance.	
3	Among Masters athletes with PVC characteristics consistent with higher risk (e.g. atypical morphology, short PVC coupling interval, repetitive PVCs, >10% burden, emergence during exercise) more extensive evaluation, including cardiac magnetic resonance imaging, ischaemic evaluation, and/or invasive electrophysiology study, should be considered.	
4	Management of Masters athletes with high-risk PVCs and/or complex ventricular arrhythmias should be based on the underlying pathology according to disease-specific guidelines.	
5	Individualized longitudinal surveillance is reasonable in Masters athletes with high-risk PVCs and no clearly defined etiology during initial evaluation.	

Author Manuscript

Author Manuscript

Author Manuscript

Author Manuscript

Table 4

Clinical consensus statements for Masters athletes with coronary atherosclerosis

#	Consideration	Strength of the advice
1	Masters athletes with established cardiovascular risk factors should receive primary preventive (cardiac) care measures including lifestyle counseling and pharmacological therapy (i.e. anti-hypertensive, lipid-lowering, etc.) for risk factor modification as dictated by general population guidelines.	
2	High cardiorespiratory fitness is associated with improved overall mortality and cardiovascular outcomes. In the absence of signs or symptoms of ischaemia, exercise habits (intensity, duration, and frequency) do not require restrictions or reductions in the presence of traditional cardiovascular risk factors, including the presence of a high coronary artery calcium score.	
3	For asymptomatic Masters athletes with low-risk cardiovascular risk profiles, routine screening with CT-derived coronary artery calcium scoring should not be performed.	
4	For asymptomatic Masters athletes deemed moderate or high-risk, including those with documented coronary calcification, further risk stratification is reasonable. Options including maximal effort-limited exercise stress testing to evaluate for silent ischaemia and/or a coronary CT angiography to evaluate the degree, location, and morphology of stenoses to inform targeted risk factor management, may be appropriate.	
5	Diagnostic ischaemic evaluation with functional testing and/or coronary CT angiography should be performed for Masters athletes with symptoms suggestive of ischaemia.	
6	Masters athletes with confirmed myocardial ischemia should receive individualized counselling on lifestyle modifications and initiation of guideline-based pharmacotherapy.	
7	Revascularization of stable coronary stenoses responsible for symptoms may be appropriate to reduce symptom burden and the potential risk of ischemic ventricular arrhythmias during high-intensity physical exertion.	
8	In the absence of randomized controlled data, revascularization of high-grade stenoses in asymptomatic Masters athletes with evidence of myocardial ischemia on functional or imaging testing may be appropriate given the potential risk of ischemic ventricular arrhythmias during high-intensity physical exertion.	
9	Similar to the general population, low-dose aspirin therapy and lipid-lowering therapy are reasonable for Masters athletes with high-risk plaque characteristics or high CACS.	
10	There is no evidence to support the initiation of prophylactic aspirin therapy prior to competition among Masters athletes without established cardiovascular risk factors.	

Author Manuscript

Author Manuscript

Author Manuscript

Author Manuscript

Table 5

Clinical consensus statements for Masters athletes with aortic dilatation

#	Consideration	Strength of the advice
1	Masters athletes with aortic dilatation (≥40 mm, irrespective of sex but after consideration of body size) should undergo a comprehensive medical evaluation, including a multigenerational family history of aortic disease and physical examination to evaluate for phenotypic traits suggestive of heritable aortic disease.	
2	Masters athletes with aortic enlargement of 45–50 mm on echocardiography should have cross-sectional imaging (computed tomography or magnetic resonance imaging) visualizing the entire aorta performed at least once. A customized imaging surveillance plan should be developed and initiated 6–12 months after index evaluation.	
3	Genetic testing is reasonable for Masters athletes with confirmed aortic enlargement of ≥45 mm.	
4	Masters athletes with aortic dimensions meeting established surgical thresholds (≥50 mm) should be counselled to avoid competitive sports and high-intensity exercise training and be referred for consideration of the timing of surgical intervention.	
5	Masters athletes with aortic dilatation measuring 40–44 mm, without a heritable etiology, are generally considered to be at low risk. Participation in competitive sports and high-intensity exercise training is reasonable, provided that appropriate clinical and imaging surveillance is maintained.	
6	Masters athletes with aortic dilatation of 45–50 mm, without a heritable etiology, are more likely at intermediate risk and require individualized evaluation. SDM should acknowledge the uncertain risks of high-intensity resistance and endurance training or competition on progression of aortic enlargement and acute aortic syndromes.	
7	Masters athletes with a history of surgical repair following aortic dissection should be counseled regarding the health benefits of low-to-moderate intensity aerobic exercise.	
8	Data defining risks of resumption of high-intensity athletic training and competition after surgical repair for aortic dissection are limited. Risks may outweigh benefits thereby necessitating SDM for Masters athletes in this context.	
9	It is reasonable for Masters athletes with BAV aortopathy who have undergone surgical aneurysm repair of the aortic root and/or ascending aorta to return to participation in all training and sporting activity with appropriate clinical and imaging surveillance.	
10	Returning to full training and sport participation after aneurysm repair in athletes with a tricuspid valve and no heritable etiology may be appropriate using a SDM model.	

Table 6 Selected studies comparing the prevalence of myocardial fibrosis in older competitive athletes.

Studies are divided by those supporting a higher prevalence of myocardial fibrosis in competitive athletes vs those that are non-supportive

Study	Subjects	Sex and Age	Training History	Sport Disciplines	LGE Findings	Mapping Findings
SUPPORTIVE						
Ragab <i>et al.</i> 2023 ¹⁰⁸	74 athletes 36 controls	75% male, 44 ± 8 years 25% female, 36 ± 7 years age- and sex-matched	10 h/week and ≥1 completed marathon race (self-reported) <3 h/week	100% marathon runners	LGE 11% athletes (63% RV insertion point; 25% non-ischemic; 12% ischemic) vs LGE 0% controls No difference males vs females; $P = .14$	↑ ECV in male runners compared to controls
Farooq <i>et al.</i> 2023 ¹⁰⁹	50 athletes 26 controls	100% male, 56 (53–64) years age-matched	>10 h/week for >15 years Compete regularly <3 h/week	82% cyclists 18% triathletes	LGE 48% athletes vs LGE 15% controls, $P = .005$ (all non-ischemic)	No difference in ECV in athletes vs controls
Domenech-Ximenes <i>et al.</i> 2020 ¹¹⁰	93 athletes 72 controls	53% male, 36 ± 6 years (total cohort) sex-matched, 34 ± 4 years	12 h/week previous 5 years <3 h/week	100% triathletes	LGE 35% athletes vs 5% controls (male, $P = .001$) LGE 41% athletes vs 0% controls (female, $P = .001$) (all hinge point)	↑ECV in athletes with LGE + vs athletes with LGE- (data available in 30%)
Tahir <i>et al.</i> 2018 ¹¹¹	83 athletes 36 controls	65% male, 44 ± 10 years 35% female, 42 ± 10 years Age- and sex-matched	>10 h/week Regular triathlons in the past 3 years <3 h/week	100% triathletes	LGE 17% athletes vs 0% controls (male, $P = .052$) (non-ischemic)	Native T1 significantly lower in triathletes vs controls (male and female) and no ECV difference
Wilson <i>et al.</i> 2011 ¹¹²	12 veteran endurance 20 veteran controls	100% male, 56 ± 6 years 100% male, 60 ± 5 years	35–52 years not specified	100% endurance	LGE 50% of veteran athletes (17% ischemic, 17% non-ischemic, 66% hinge point) LGE 0% in young and veteran controls	-
Breuckmann <i>et al.</i> 2009 ¹¹³	102 runners 102 controls	100% male, 57 ± 6 years Age- and sex-matched	completed ≥5 marathons in the past 3 years not specified	100% marathon runners	LGE 12% athletes vs 4% controls ($P = .077$) (42% ischemic 58% non-ischemic)	-
NON-SUPPORTIVE						
Andresen <i>et al.</i> 2024 ¹¹⁴	27 athletes 16 controls	100% male, 41 ± 9 years Age- and sex-matched	>24 MET/week for the past 12 years not specified	marathon, triathlon, cycling (elite/Olympic)	LGE 12% athletes vs 0% controls ($P = .14$) (non-ischemic)	No difference in ECV between athletes vs controls
Missenard <i>et al.</i> 2021 ¹¹⁵	33 athletes 18 controls	94% male, 47 ± 6 years Age- and sex-matched	Minimum 8 h/week in the past 15 years Untrained individuals among military forces	Running, triathlon, cycling	LGE 5% controls (inferior wall) vs 0% athletes ($P = .33$)	No difference in T1 mapping/ECV between athletes vs controls
Maleket <i>et al.</i> 2019 ¹¹⁶	30 athletes 10 controls	100% male, 41 ± 7 years Age- and sex-matched	70 km/week running Healthcare personnel	100% ultramarathon runners	LGE 27% athletes vs 10% controls ($P = .4$) (62.5% RV insertion point, 37.5% non-ischemic)	No difference in ECV between athletes vs controls
Pujadas <i>et al.</i> 2018 ¹¹⁷	34 athletes 12 controls	100% male, 48 ± 7 years Age- and sex-matched	>10 years of training marathon time <3 h Untrained individuals	100% runners	LGE 9% athletes vs 0% controls (non-ischemic) no correlation between total training volume and LGE	No difference in T1 mapping/ECV between athletes vs controls
Abdullah <i>et al.</i> 2016 ¹¹⁸	21 athletes 71 controls	73% male, >65 years Age- and sex-matched	6–7 exercise sessions/week, 25 years (number of controls based on training session/week): –25 with 1 session –23	marathon and triathlon	LGE 0% athletes vs 1.5% controls (hinge-point)	-

Author Manuscript

Author Manuscript

Author Manuscript

Author Manuscript

Study	Subjects	Sex and Age	Training History	Sport Disciplines	LGE Findings	Mapping Findings
Bohm et al, 2016 ¹¹⁹	33 athletes of whom 16 former world-class athletes 33 controls	100% male, 30–60 years Age- and sex-matched	with 2–3 sessions –23 with 4 + sessions ≥ 10 h/week ≥10 years ≤3 h/ week	Running, triathlon, cycling	LGE 3% athletes vs 0% controls (non-ischemic)	-

Table 7
Clinical consensus statements for Masters athletes with myocardial fibrosis

#	Consideration	Strength of the advice
1	In Masters athletes with myocardial fibrosis identified after CMR with LGE, further clinical investigation(s) should be based on historical features, particularly the presence of arrhythmic symptoms and/or ventricular arrhythmias, presence of underlying cardiovascular risk factors, supportive findings on other diagnostic tests (e.g. 12-lead ECG), and the pattern and extent of LGE.	
2	Clinical treatment(s) for myocardial fibrosis in Masters athletes should be based on the underlying clinical diagnosis and clinical risk stratification to lower risk for sudden cardiac arrest/death with treatment of ventricular arrhythmias and/or reduce symptoms.	
3	Asymptomatic Masters athletes with isolated LGE by CMR at the right ventricular insertion/hinge point(s) do not require further evaluation and/or clinical risk stratification.	
4	Among asymptomatic Masters athletes with incidentally detected LGE detected by CMR in a pattern associated with sudden cardiac arrest/death risk (e.g. isolated non-ischaemic left ventricular LGE with a stria pattern), it is reasonable to proceed with additional clinical risk stratification with maximal effort and sport-specific exercise testing, ambulatory rhythm monitoring (including during training), and serial clinical surveillance to rule out complex ventricular arrhythmias.	
5	The determination of competitive sports and/or intense physical activity for Masters athletes with myocardial fibrosis should incorporate shared decision-making and consider the underlying clinical diagnosis, pattern and extent of LGE, and results of additional clinical risk stratification, particularly the presence of ventricular arrhythmias.	

Table 8
Clinical consensus statements for Masters athletes with unexplained cardiomyopathy

#	Consideration	Strength of the advice
1	Exercise-induced cardiomyopathy, a proposed heart muscle disease attributed to longstanding exercise training with no other apparent etiology, is a diagnosis of exclusion. To date, no formal diagnostic criteria have been established for this condition; however, its proposed features include disproportionate RV dilation, RV systolic dysfunction, and ventricular arrhythmias of RV origin.	
2	Masters athletes with an extensive endurance sports history presenting with unexplained cardiomyopathy, characterized by RV dilatation out of proportion to LV dilatation and ventricular arrhythmias of RV origin, require a comprehensive diagnostic evaluation that includes contrast-enhanced CMR, genetic testing, rhythm monitoring, and exercise testing to exclude other forms of cardiomyopathy.	
3	The role of an invasive diagnostic electrophysiology study in Masters athletes with unexplained cardiomyopathy and symptomatic ventricular arrhythmias is uncertain but may be appropriate to guide risk stratification and identify athletes who may benefit from ablation.	
4	Among Masters athletes with cardiomyopathy of unexplained etiology, shared decision-making is suggested to discuss management decisions on sports participation, prescribed detraining, ongoing evaluation and management of ventricular arrhythmias, risk stratification to evaluate the need for ICD implantation, and the frequency of serial imaging to monitor disease progression.	